

AT32 Motor Monitor Application Note

Introduction

This application note mainly introduces how to use AT32 MCU with motor control library and PC software, how to debug control parameters and how to control the motor, helping users to have a better understanding of PC software features, operation methods and usage precautions, and use PC software for debugging of motor parameters and control parameters.

Applicable products

Part number	AT32Fxx, AT32Lxx
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1 Software and hardware requirements

A BLDC motor, AT-Link or J-Link, AT motor development board and AT motor control library are required.

Run the executable “ArteryMotorMonitor.exe” directly, without the need for installation.

1.1 Hardware

- Windows®-based PC (Windows 8, Windows 10, Windows 11) to install user control interface program
- One micro-B USB cable to connect the development board with PC for communication
- ARTERY AT-Link or 3rd-party programmer
- A 3-phase AC motor with 12V~60V rated voltage and below 30A rated current
- DC power supply
- ARTERY motor development board

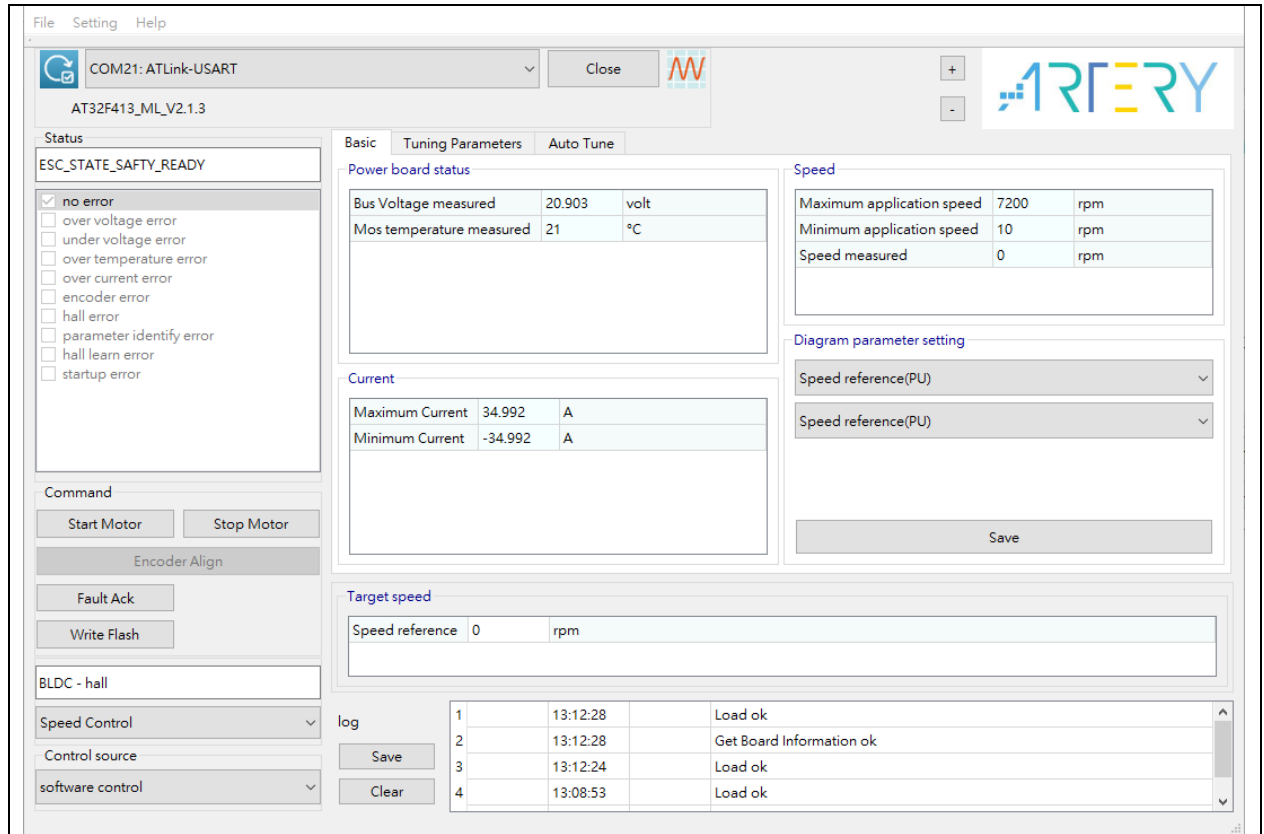
1.2 Software

- ARTERY AT32 motor control demonstration project program
- Keil® µvision IDE (µvision V5.36.0.0 is used in this example)

2 User interface operation

Figure 1 below shows the user interface of ARTERY Motor Monitor, and related function blocks are detailed in subsequent sections.

Figure 1. ARTERY Motor Monitor user interface



2.1 Connection settings

After the hardware and software are well prepared, set up connection between UI and the control board as follows:

STEP-1

Connect the motor, AT-Link/J-Link and power supply to the motor development board, and connect USART interface to PC via an USB cable.

STEP-2

Use MDK to compile demo project code, and use J-Link or AT-Link to download to the on-board chip.

STEP-3

Run ArteryMotorMonitor_V2.1.x.exe (software version: V2.1.x); click File -> Open Project and select ArteryMotorMonitor_V2.1.x.atmcx-> Open.

STEP-4

Click the Serial Port icon (1) and select the corresponding serial port (2); then click Open (3) to enable real-time communication, as shown in Figure 2 below.

Figure 2. Connection setting



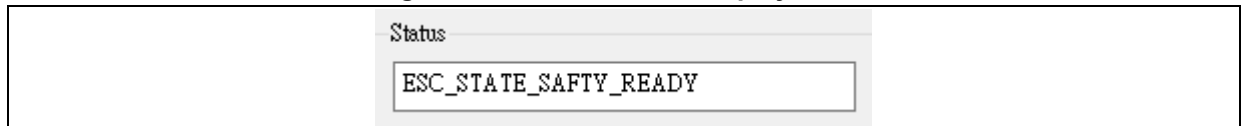
2.2 Control and status display list

This region contains the display area of status machine and error type, and the operation area of control command button and droplist of control mode to perform motor startup/stop, encoder alignment, writing parameters to Flash, etc., or switch control mode, such as open loop control, voltage control, ID/IQ tune, torque control, speed control and position control.

2.2.1 Status machine

It displays the current status of motor control program, including Idle, Safety ready, Angle init, Starting, Running, Free run, I_tune, Enc_align, Winding param Id, Auto learn and Error.

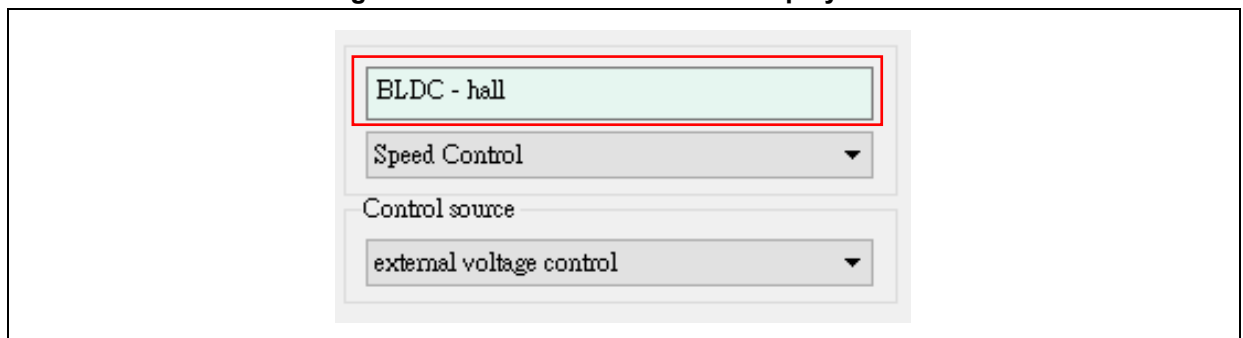
Figure 3. Status machine display area



2.2.2 Firmware control mode

It displays the firmware control mode, as shown in Figure 4. In this example, it is the six-step square-wave control mode with Hall sensor.

Figure 4. Firmware control mode display area



2.2.3 Error type

It displays the type of error in motor running process, including over-voltage, under-voltage, over-temperature, over-current, encoder error, Hall error, parameter identify error, Hall learn error and startup error.

Figure 5. Error type display area



2.2.4 Control command button operation

- 1) This application software contains five control command buttons that are used for motor startup/stop, encoder alignment, write parameters to Flash, error clearing, etc.

Figure 6. Control command buttons



- 2) Start Motor
Click to start up motor.
- 3) Stop Motor
Click to stop motor.
- 4) Encoder Align
Click to perform encoder alignment to zero position (invalid in open loop control mode).
- 5) Write Flash
When parameters have been debugged or in case of modifying motor control parameters, click to write parameters to Flash. After this command is executed, the Flash will store these parameters so that users do not need to re-debug motor control parameters for the next power-on.

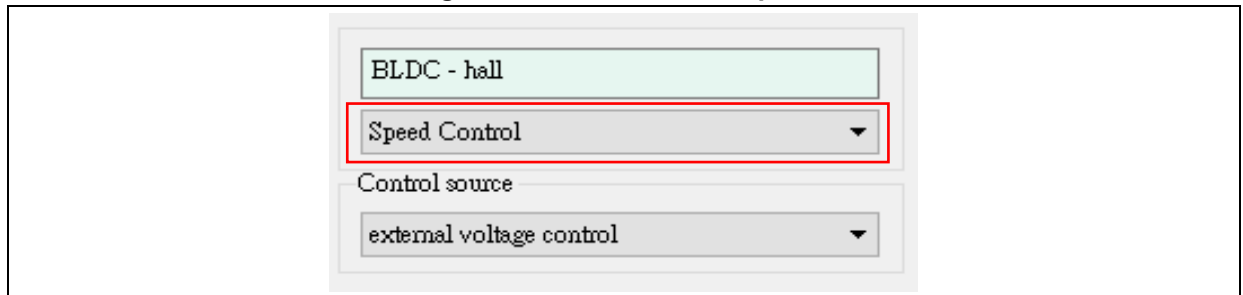
6) Fault Ack

Click to clear the current error status.

2.2.5 Control mode droplist

Totally seven control modes are optional, including open loop control, voltage control, ID/IQ tune, torque control, speed control and position control, and parameters are different in different modes. Therefore, this area should be used together with the parameter setting page, which is detailed in Section 2.4.

Figure 7. Control mode droplist



2.3 Basic parameters

The basic parameters display/setting page contains Vdc voltage and MOS temperature monitoring, maximum/minimum application speed, maximum/minimum current, maximum/minimum application angle, target angle/speed/torque, and drawing area channel selection.

Figure 8. Basic parameters setting page

2.3.1 Voltage and temperature display

It displays the real-time voltage (unit: Volt) and MOS temperature (unit: °C) to monitor and check for over-voltage, under-voltage or over-temperature. In case of any error, it will be displayed in the error list as mentioned in Section 2.2.3.

Figure 9. Voltage and temperature display area

2.3.2 Maximum/minimum speed

The maximum/minimum speed and current speed (unit: rpm) are displayed in this area. This interface only supports read access, and modification to the maximum/minimum speed should be performed in firmware, for example, modifying the MAX_SPEED_RPM and MIN_SPEED_RPM in *motor_control_drive_param.h* file.

Figure 10. Maximum/minimum/current speed display area

Maximum application speed	4200	rpm
Minimum application speed	10	rpm
Speed measured	0	rpm

Figure 11. Define the maximum/minimum speed

```

63 #define MIN_SPEED_RPM          (10)
64 #define MAX_SPEED_RPM          (4200)
65 #define MIN_CONTROL_SPEED      (120)

```

2.3.3 Maximum/minimum current

The maximum/minimum current (unit: ampere) for motor control can be read and set in this area. The current can be adjusted according to the motor characteristics or the driver board. By default, the maximum/minimum current set by firmware is read. To adjust the current, switch to ID Tune or IQ Tune mode and then double click to change the value, and the bottom LOG displays a message after successful setting.

Figure 12. Maximum/minimum current display area

Maximum Current	4.98962	A
Minimum Current	0.997925	A

Figure 13. LOG message after successful current setting

	Time	Motor	Message
1	17:28:59		Set REG 36 = 3932.16:OK
2	17:28:52		Set REG 36 = 3276.8:OK

2.3.4 Maximum/minimum application angle (PMSM encoder/Hall sensor)

The maximum/minimum application angles (unit: degree) for motor control are displayed in this area. This interface only supports read access, and modification to the maximum/minimum application angles should be performed in firmware, for example, modifying the MAX_POSITION_ANGLE and MIN_POSITION_ANGLE in *motor_control_drive_param.h* file.

Figure 14. Maximum/minimum application angle display area

Angle		
Max application angle	36000	degree
Min application angle	-36000	degree

Figure 15. Define the maximum/minimum application angle

```

56 #define MAX_POSITION_ANGLE      36000    /* Degree */
57 #define MIN_POSITION_ANGLE     (-MAX_POSITION_ANGLE)

```

2.3.5 Target speed/torque setting

Different control parameters are set for different control sources. For example, the “Target speed” is displayed in Speed Control mode, as shown in Figure 16, and the “Torque reference” is displayed in Torque Control mode, as shown in Figure 17.

The target speed is communicated in rpm, and the torque reference is communicated in amperes. There are two available control sources, including the external control and software control.

1) External voltage control source

This application software supports external control source. Open the droplist of “Control source” and select “external voltage control”, so that users can adjust the speed or torque reference via external voltage source. The speed or torque reference field displays the converted speed or torque by external voltage source.

Note: This field cannot be modified in the external control source mode

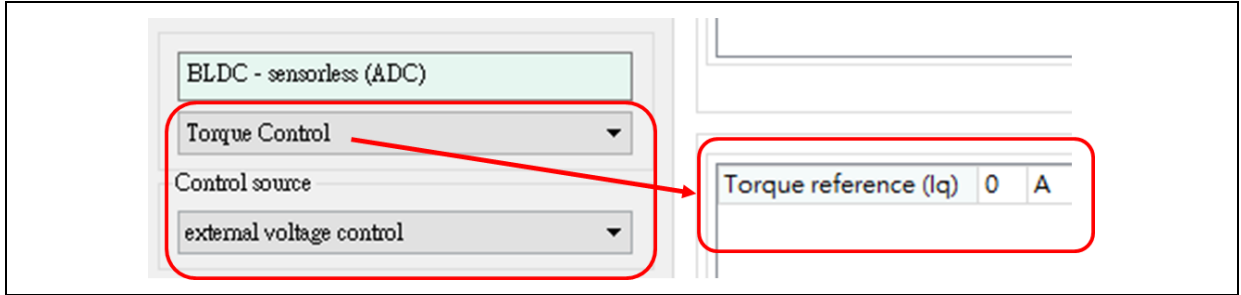
Figure 16. Speed control mode (external voltage control source)

The screenshot shows the application interface with the following settings:

- Motor type: BLDC - sensorless (ADC)
- Control mode: Speed Control
- Control source: external voltage control
- Target speed: Speed reference 0 rpm

A red box highlights the 'Speed Control' and 'external voltage control' dropdown menus, and another red box highlights the 'Target speed' field. A red arrow points from the 'Speed Control' dropdown to the 'Target speed' field.

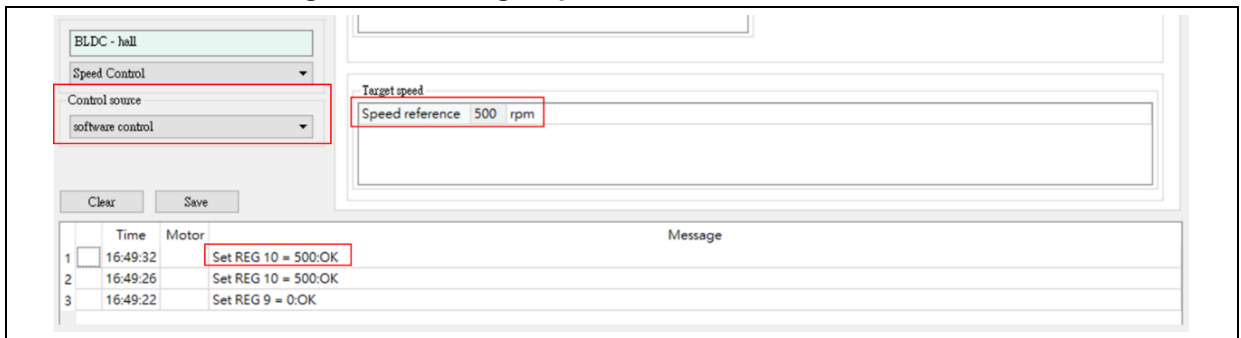
Figure 17. Torque control mode (external voltage control source)



2) Software control source

The software control mode is selected, by default. Open the droplist of “Control source” and select “Software control”, as shown in Figure 18. In this mode, users can change the speed/torque reference (double click this field to change the value) in UI interface to adjust the target speed/torque in motor control. The bottom field displays a message after successful setting.

Figure 18. Set target speed in software control mode



2.3.6 Position reference/position measured (encoder mode)

The position reference and position measured are displayed when the position control mode is selected, as shown in Figure 19. The position reference is communicated in degrees. In position control mode, users can change the position reference (double click this field to change the value) in UI interface to adjust the motor rotor position. The bottom field displays a message after successful setting.

Figure 19. Position reference and position measured



2.3.7 Waveform drawing and parameter setting

This application supports dual-channel waveform drawing, as shown in Figure 20. The droplist “Ia” and “Ib” are used to select channel 1 and channel 2 parameters, respectively. Select the required parameters and then click “Save”. Click the drawing icon as shown in Figure 21 to generate a new window and draw the waveform.

Figure 20. Set drawing channel parameters

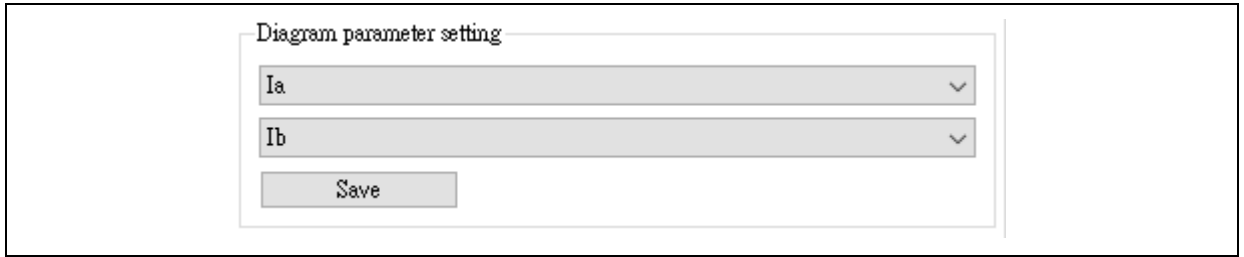


Figure 21. Waveform drawing icon



As shown in Figure 22, the left vertical axis represents channel 1 and the right represents channel 2. The waveform color is the same as that of the axis. For example, channel 1 is represented by "Ia (A)", and the corresponding waveform is drawn in green; channel 2 is represented by "Ib (A)", and the corresponding waveform is drawn in purple. Besides, the sampling rate of the waveform is correlated to the PWM interrupt frequency. If the PWM interrupt frequency is 20 KHz, the sampling period is 50 us (maximum 32s can be recorded).

Figure 22. Waveform drawing

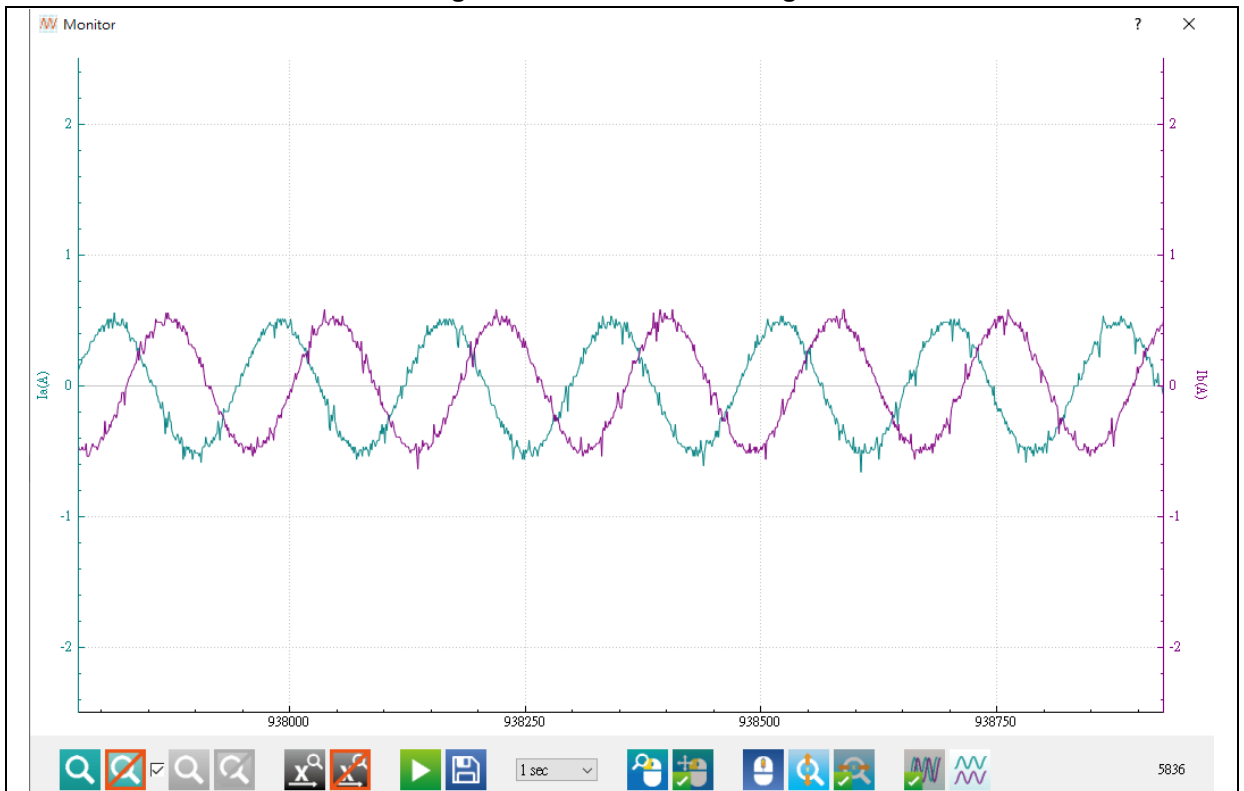


Table 1 lists the waveform drawing buttons.

Table 1. Waveform drawing buttons

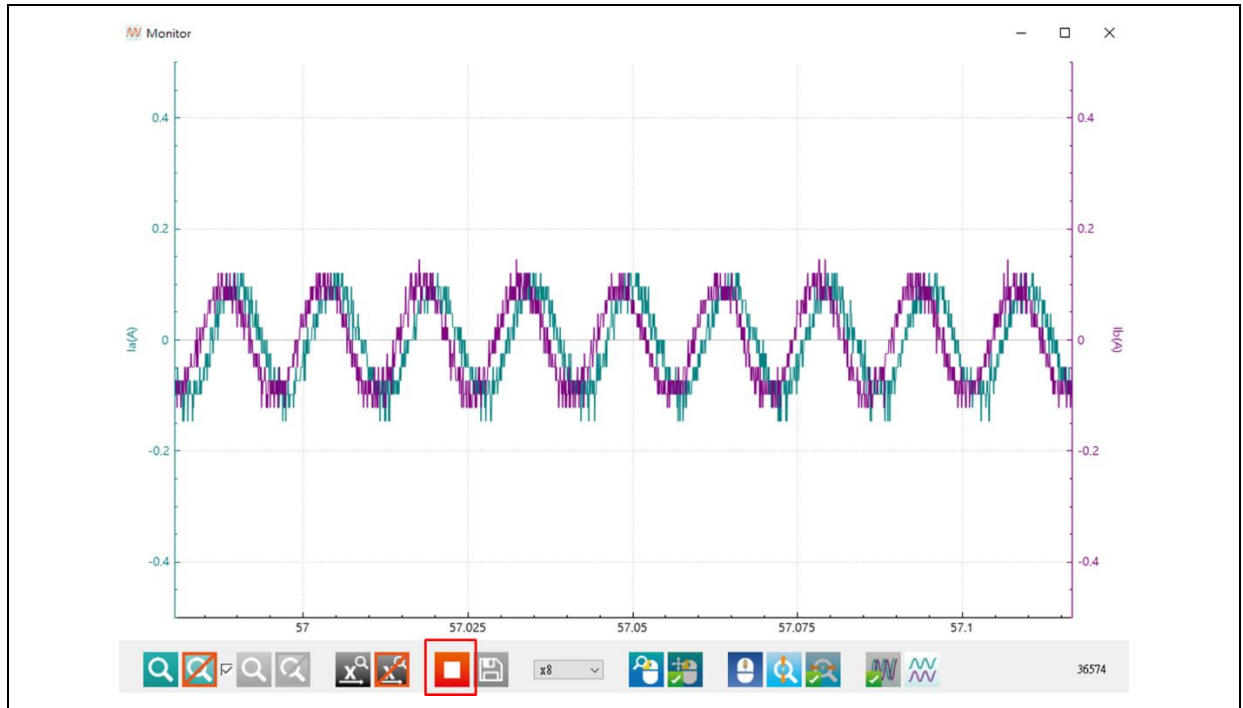
Icon	Description
	Set the display range of channel 1 vertical coordinate
	Disable zoom-in (channel 1 vertical coordinate)
	Set the display range of channel 2 vertical coordinate
	Disable zoom-in (channel 2 vertical coordinate)
	Tick to synchronize channel 1 and channel 2 coordinates
	Untick to cancel synchronization of channel 1 and channel 2 coordinates
	Set the display range of horizontal coordinate
	Disable horizontal axis zoom-in
	Start waveform update
	Pause waveform update
	Save the current waveform and data (after waveform update is paused)
	Display range of waveform horizontal coordinate (drop-down menu: 1s, 2s, 4s, 8s, 16s and 32s)
	Right click to zoom in
	Right click to move up/down and left/right
	Switch mouse wheel functions (zoom in/out, or move horizontally/vertically)
	Slide the mouse wheel to zoom in/out the horizontal and vertical axes simultaneously
	Slide the mouse wheel to zoom in/out the vertical axis
	Slide the mouse wheel to zoom in/out the horizontal axis
	No reaction when slide the mouse wheel
	Slide the mouse wheel to move the horizontal and vertical axes simultaneously
	Slide the mouse wheel to move the vertical axis
	Slide the mouse wheel to move the horizontal axis
	No reaction when slide the mouse wheel
	Waveforms of channel 1 and channel 2 are displayed on the same window
	Waveforms of channel 1 and channel 2 are displayed on different windows

2.3.8 Waveform storage

The application provides the function of storing waveforms as images or data, the operation steps are as follows:

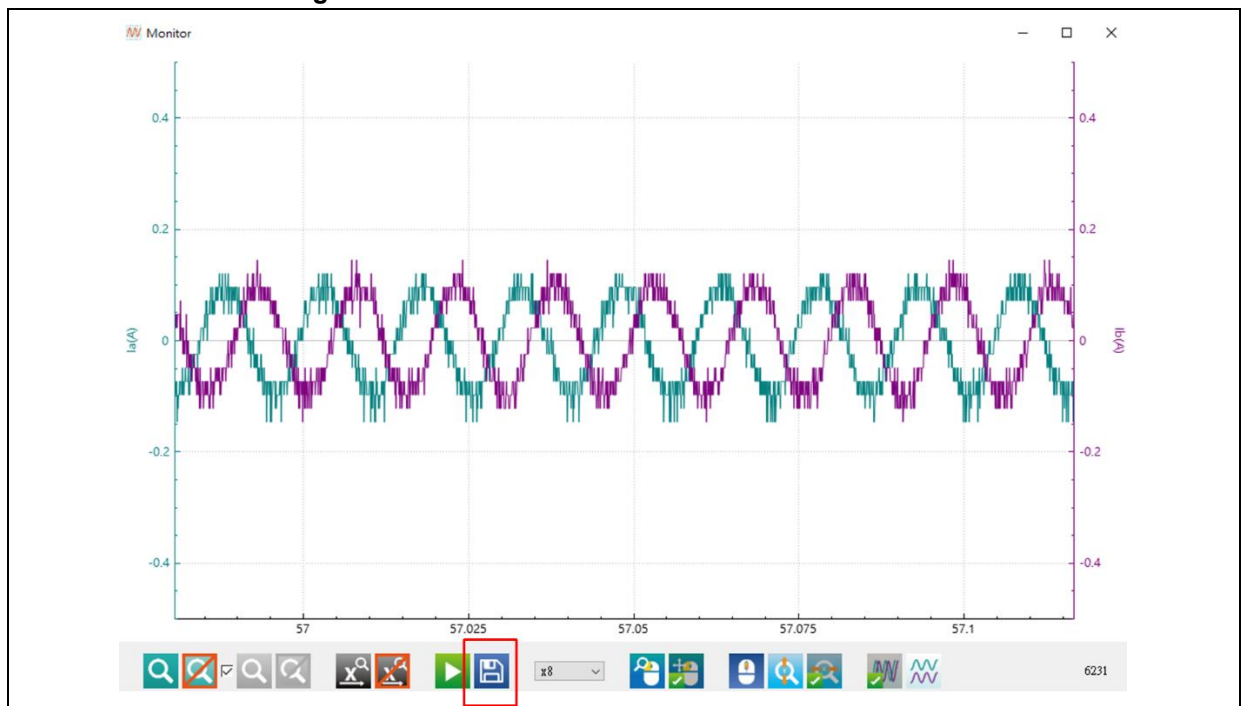
STEP-1. Click the  button to pause waveform update.

Figure 23. Click the pause waveform update button



STEP-2. Click the  button to save waveform and data.

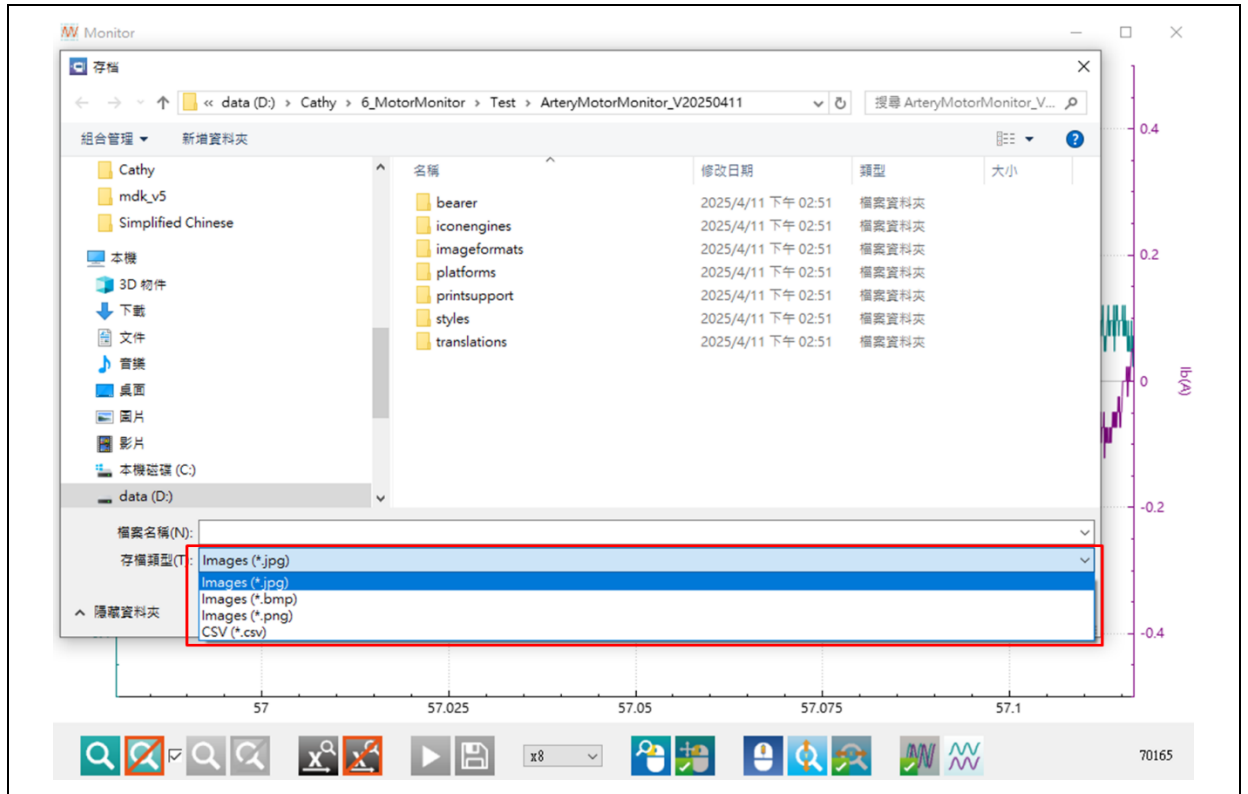
Figure 24. Click the save waveform and data button



STEP-3. Drop down to select the file format for storage, select (*.jpg/*.bmp/*.png) for storing

pictures, select *.csv for storing data, and enter the file name you want to store to finish.

Figure 25. Selecting a file extension



2.4 Parameter setting

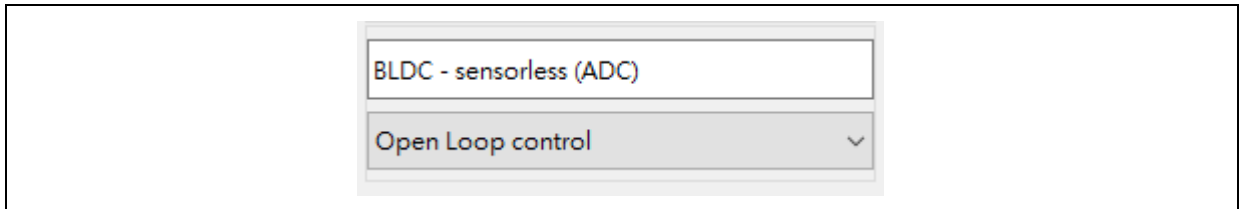
It is an advanced parameter setting page. Users can refer to Section 2.2.5 to modify parameters on this page, and start motor to confirm response or click drawing icon to view the response waveforms after debugging. For details about waveform drawing, please refer to Section 2.3.7.

2.4.1 Six-step square wave open loop control

Select open loop control option to verify that the motor is running properly. This step helps to ensure that the basic control mechanism is functioning before moving to more complex control methods. The open loop voltage and speed are incremented according to the motor running speed and motor phase current value.

STEP-1: Select "Open Loop control".

Figure 26. Select open loop control



STEP-2: Go to "Tuning Parameters" (as shown in Figure 27) and set parameters in OpenLoop initial voltage, OpenLoop initial speed, OpenLoop final speed, OpenLoop times, and OpenLoop increase volt. Figure 28 shows the open loop control.

Figure 27. Tuning open loop control parameters (BLDC)

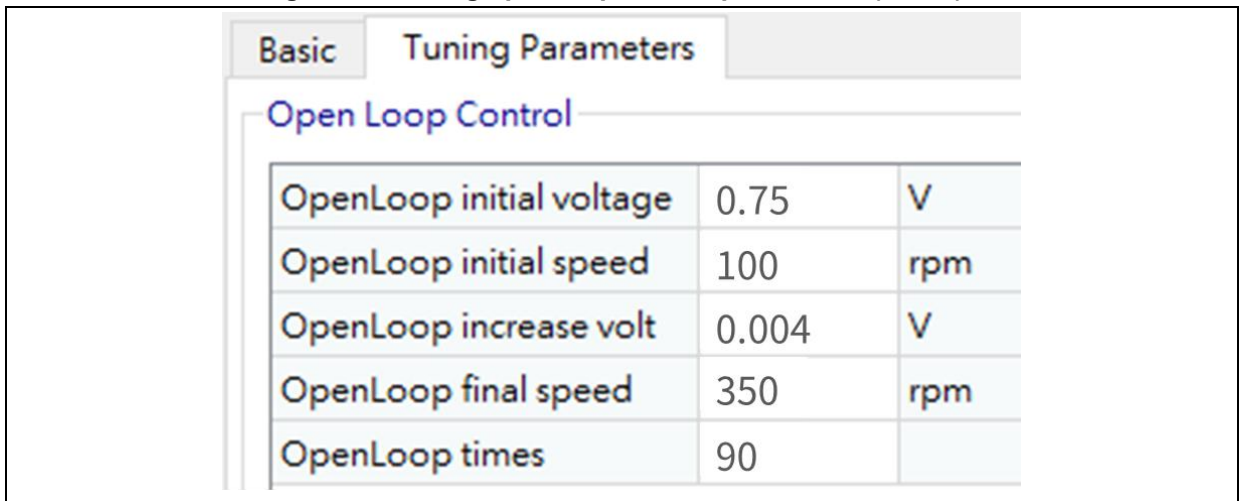
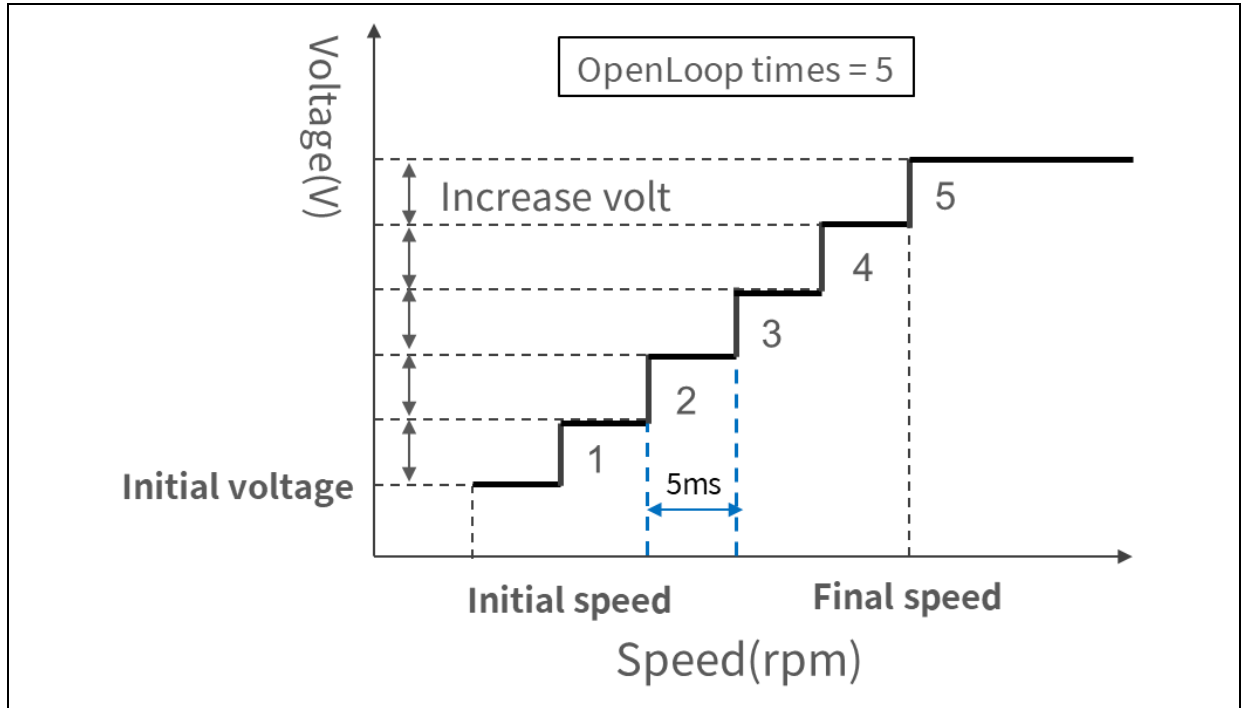


Figure 28. Schematic diagram of open loop control



Tips:

1. Initially, set **Openloop increase volt** and **Openloop times** to 0, and set **Openloop final speed** and **Openloop initial speed** to the same value. Then find the appropriate starting voltage and speed.
2. Set the final speed according to the initial speed, and adjust **increase volt** and **Openloop times**. In case of underpower, increase **increase volt** value or **Openloop times** value (it is recommended to be >50).

STEP-3: Click on “Start Motor”, and monitor the current until the motor starts to run properly (set the open loop voltage value appropriately to prevent overheating damage to the motor).

STEP-4. Fill in the parameters into macro definitions in the *motor_control_drive_param.h* file, as shown in Figure 29, and recompile and re-program the code.

Figure 29. Macro definition of open loop control parameters

```

motor_control_drive_param.h
225 /* open loop control */
226 #define OLC_INIT_SPD          (100)      /*!< rpm */
227 #define OLC_FINAL_SPD        (350)      /*!< rpm */
228 #define OLC_TIMES             (90)       /*!< Accumula
229 #define OLC_INIT_VOLT         (0.75)     /*!< V */
230 #define OLC_VOLT_INC          (0.004)

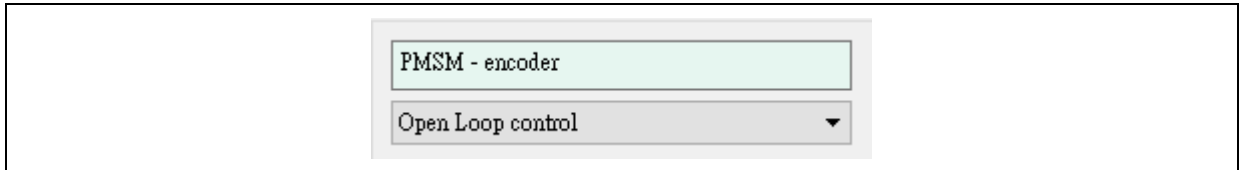
```

2.4.2 Open loop control (FOC control)

Select “Open loop voltage” in the open loop control mode to drive the motor without position sensor, and check whether the motor runs properly and whether the running direction is correct. The encoder also can be used to check whether the running direction in encoder mode is correct. In sensorless FOC mode, this mode is also used for preliminary adjustment of observer parameters. The open loop voltage and open loop angle increment are increased from 0 according to the motor running speed and phase current.

STEP-1: Select “Open Loop control”.

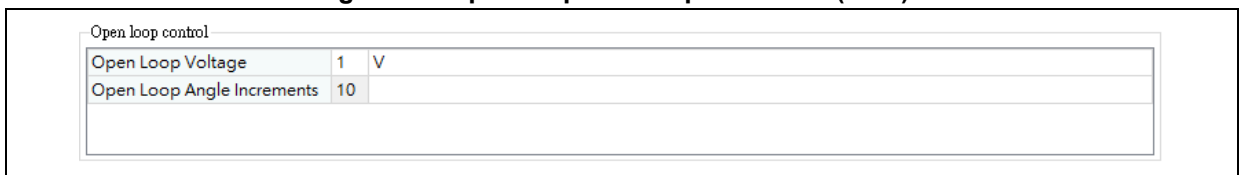
Figure 30. Select open loop control (FOC)



The screenshot shows a control interface with two main components. The top component is a green rectangular box containing the text "PMSM - encoder". Below it is a grey rectangular box containing a dropdown menu with the text "Open Loop control" and a downward-pointing arrow.

STEP-2: Slowly increase “Open Loop Voltage” and “Open Loop Angle Increments” values and observe the phase current. Make sure that the motor runs smoothly (excessive open loop voltage may cause overheating damage to motor).

Figure 31. Open loop control parameters (FOC)



The screenshot shows a table titled "Open loop control" with two rows of parameters. The first row is "Open Loop Voltage" with a value of "1" and a unit of "V". The second row is "Open Loop Angle Increments" with a value of "10".

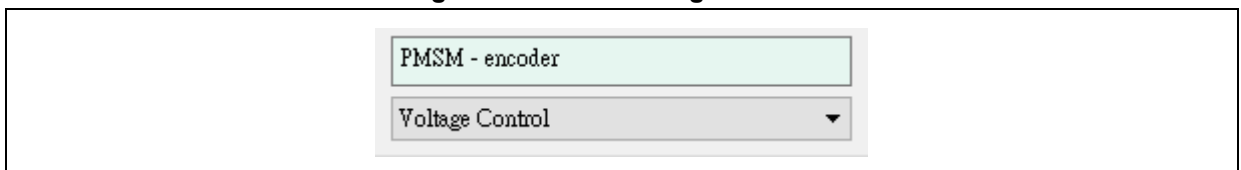
Open loop control		
Open Loop Voltage	1	V
Open Loop Angle Increments	10	

2.4.3 Voltage control

Based on the position sensor, the voltage control mode can be selected to control motor D/Q axis voltage.

STEP-1: Select “Voltage Control”.

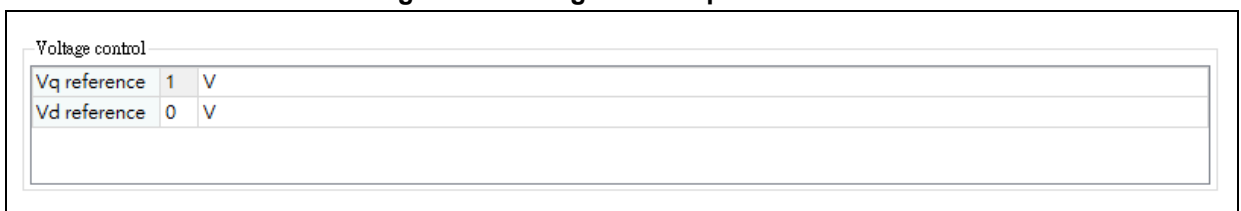
Figure 32. Select voltage control



The screenshot shows a control interface with two main components. The top component is a green rectangular box containing the text "PMSM - encoder". Below it is a grey rectangular box containing a dropdown menu with the text "Voltage Control" and a downward-pointing arrow.

STEP-2: Adjust the Q axis voltage to drive the motor to run; set the D axis voltage to position the motor rotor pole to D axis.

Figure 33. Voltage control parameters



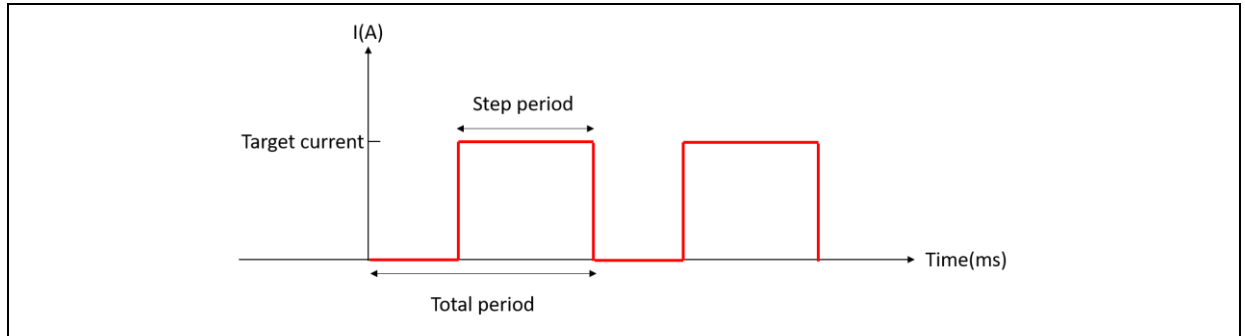
The screenshot shows a table titled "Voltage control" with two rows of parameters. The first row is "Vq reference" with a value of "1" and a unit of "V". The second row is "Vd reference" with a value of "0" and a unit of "V".

Voltage control		
Vq reference	1	V
Vd reference	0	V

2.4.4 ID tune

A step current is generated in ID Tune mode, as shown in Figure 34. Parameters related to the step current are adjustable. The step current is generated to help check the current response after ID PID parameters adjustment.

Figure 34. Schematic diagram of step current



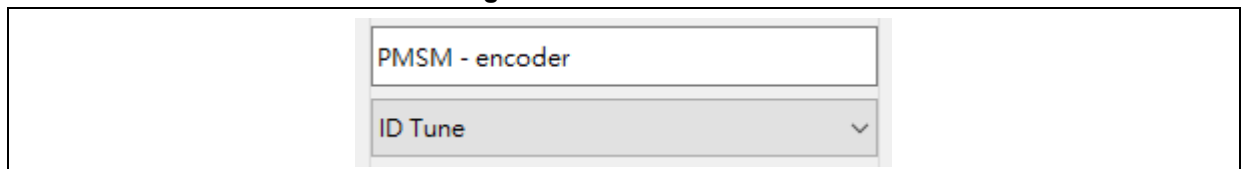
Note:

1. The ID tuning feature is not available for six-step square-wave control.
2. In FOC control mode, it is recommended to tune ID and then IQ. Before tuning current, set KP and KI of D and Q axes to 0.

Follow the below steps.

STEP-1: Select "ID tune".

Figure 35. Select ID tune



STEP-2: Go to "Tuning Parameters" window, and set PID parameters and step current parameters. Parameters obtained by current PI controller parameters auto tuning can be used for initial adjustment, and users can adjusted these parameters as needed to get the desired current response, as shown in Figure 36.

Figure 36. PID parameters and step current parameters

Basic			Tuning Parameters		
Unit step config			ID Current control		
Current Tune target current	0.999	A	Flux KP	25000	
Current Tune total period	100	ms	Flux KI	3000	
Current Tune step period	2	ms	Flux KP DIV	2048	
			Flux KI DIV	4096	

STEP-3: Click on "Start Motor".

STEP-4: Set the “Flux reference(I_d)” and “Flux measured(I_d)” in “Diagram parameter setting”, and then click “Save”.

Figure 37. Modify channel monitoring parameters (ID tune)

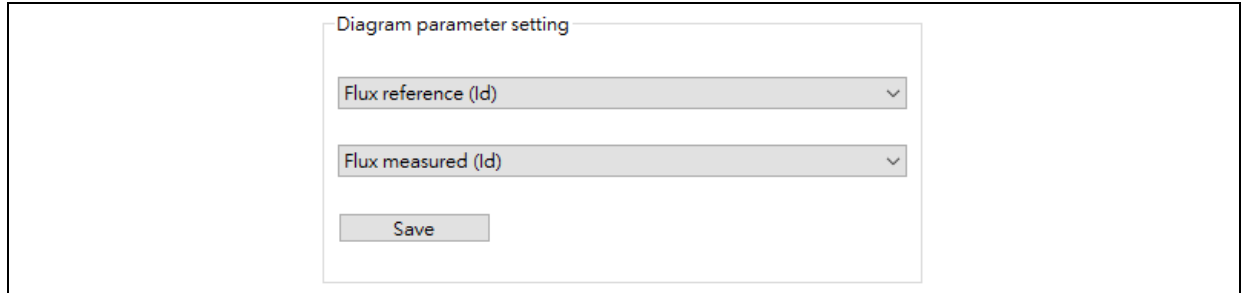


Diagram parameter setting

Flux reference (Id) ▼

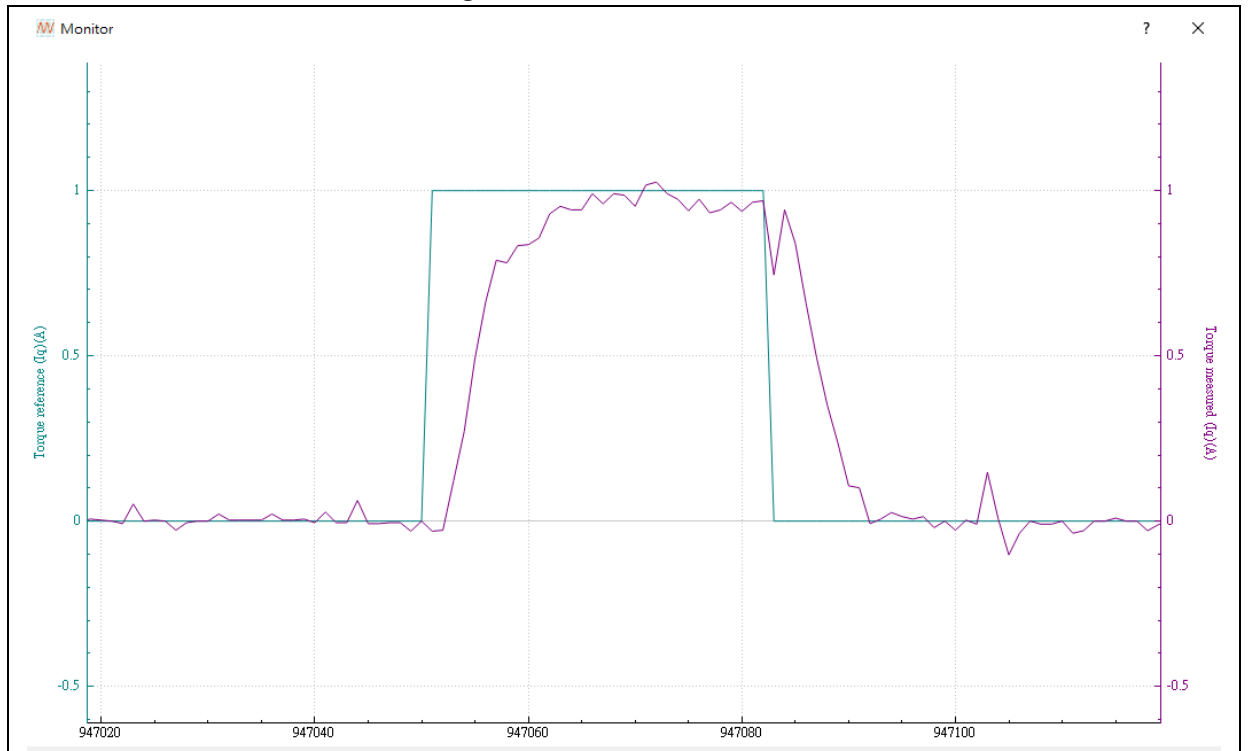
Flux measured (Id) ▼

Save

STEP-5: Click the drawing icon to open the waveform window.

STEP-6: Check whether the current response is as expected, as shown in Figure 38. If it is not as expected, click to stop the motor and repeat STEP-2~STEP-6.

Figure 38. ID tune waveform



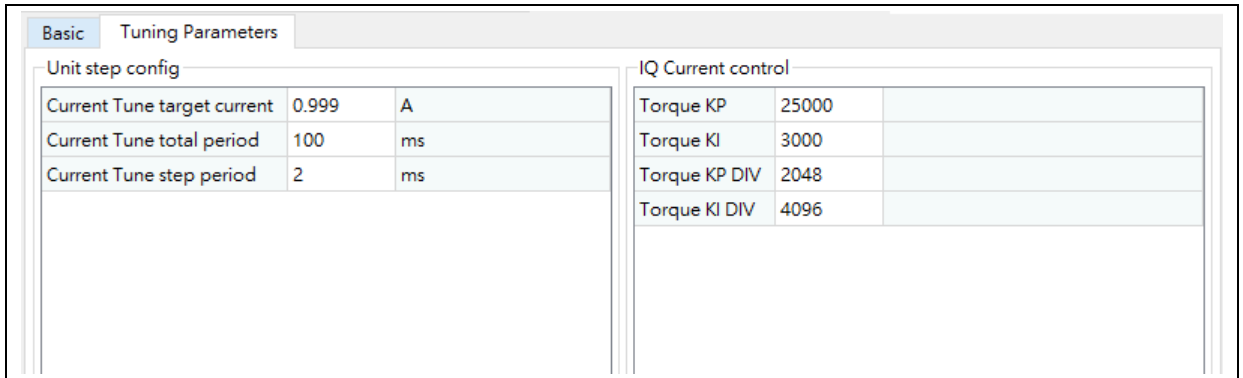
2.4.5 IQ tune

A step current is generated in IQ Tune mode, as shown in Figure 34. Parameters related to the step current are adjustable. The step current is generated to help check the current response after IQ PID parameters adjustment.

STEP-1: Select "IQ tune".

STEP-2: Go to "Tuning Parameters" window and set PID parameters and step current parameters. Parameters obtained by current PI controller parameters auto tuning can be used for initial adjustment, and users can adjust these parameters as needed to get the desired current response.

Figure 39. Q axis PID and step current related parameters



Unit step config			IQ Current control		
Current Tune target current	0.999	A	Torque KP	25000	
Current Tune total period	100	ms	Torque KI	3000	
Current Tune step period	2	ms	Torque KP DIV	2048	
			Torque KI DIV	4096	

STEP-3: Click on "Start Motor".

STEP-4: Set the "Torque reference(Iq)" and "Torque measured(Iq)" in "Diagram parameter setting", and then click "Save".

Figure 40. Modify channel monitoring parameters (IQ tune)

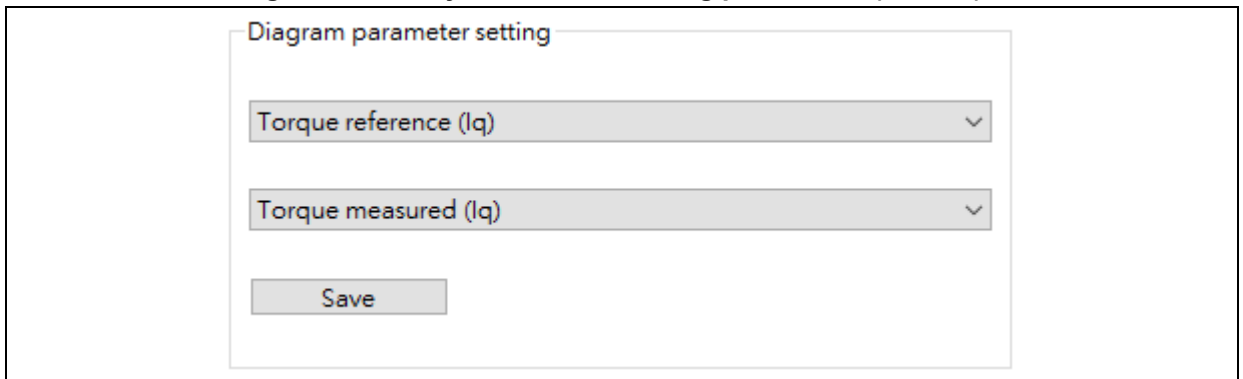


Diagram parameter setting

Torque reference (Iq) ▼

Torque measured (Iq) ▼

Save

STEP-5: Click the drawing icon to open the waveform window.

STEP-6: Check whether the current response is as expected, as shown in Figure 38. If it is not as expected, click to stop the motor and repeat STEP-2~STEP-6.

2.4.6 Current control

In current control mode, users can adjust the flux reference to control the torque, and check the response through waveform drawing.

STEP-1: Select "Torque Control".

STEP-2: Select "Software control" in the "Control source" and then set the "Torque reference"

Figure 41. Set torque/flux reference

Torque reference (Iq)	0.100	A
Flux reference (Id)	0.000	A

STEP-3: Click on "Start Motor".

STEP-4: Set the "Ia" and "Ib" in "Diagram parameter setting", and then click "Save".

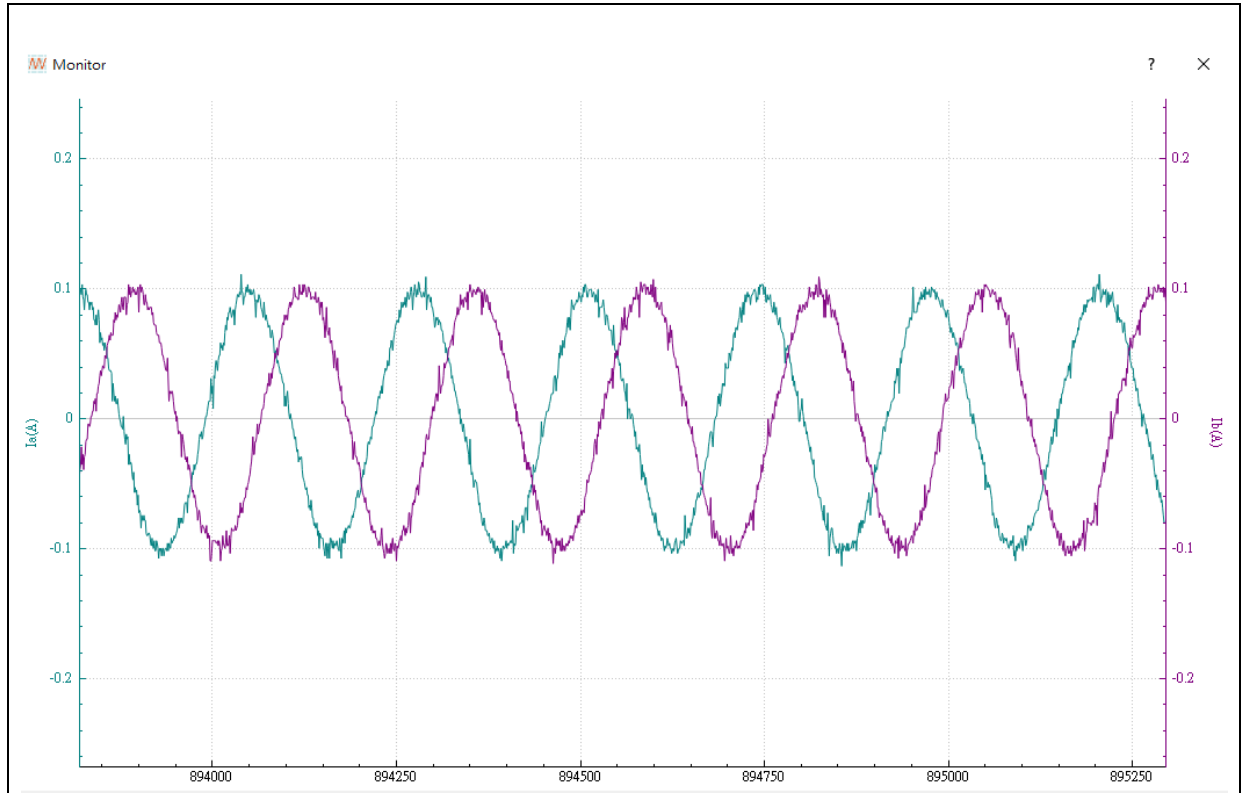
Figure 42. Modify channel monitoring parameters (current control)

Diagram parameter setting	
Ia	▼
Ib	▼
Save	

STEP-5: Click the drawing icon to open the waveform window.

STEP-6: Check whether the current waveform is as expected, as shown in Figure 43.

Figure 43. Waveform in current control mode



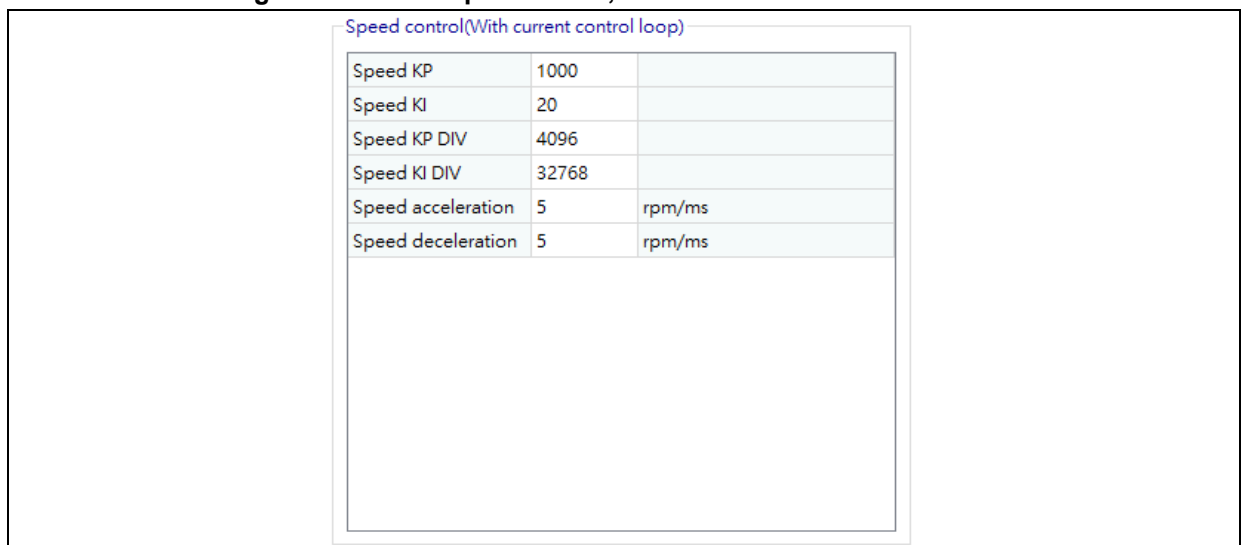
2.4.7 Speed control

In speed control mode, users can adjust the speed PID parameters, acceleration and deceleration, and check the response after adjustment through waveform drawing

STEP-1: Select “Speed Control”.

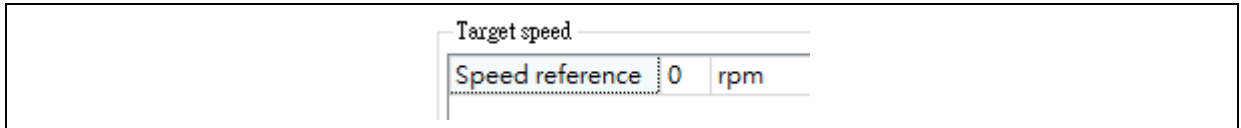
STEP-2: Set speed PID parameters, acceleration and deceleration.

Figure 44. Set PID parameters, acceleration and deceleration



STEP-3: Select “Software control” in “Control source” and set the target speed (Speed reference).

Figure 45. Set target speed



Target speed

Speed reference 0 rpm

STEP-4: Click on “Start Motor”.

STEP-5: Set “Speed reference(PU)” and “Speed measured(PU)” in “Diagram parameter setting”, and then click “Save”.

Figure 46. Modify channel monitoring parameters (speed control)

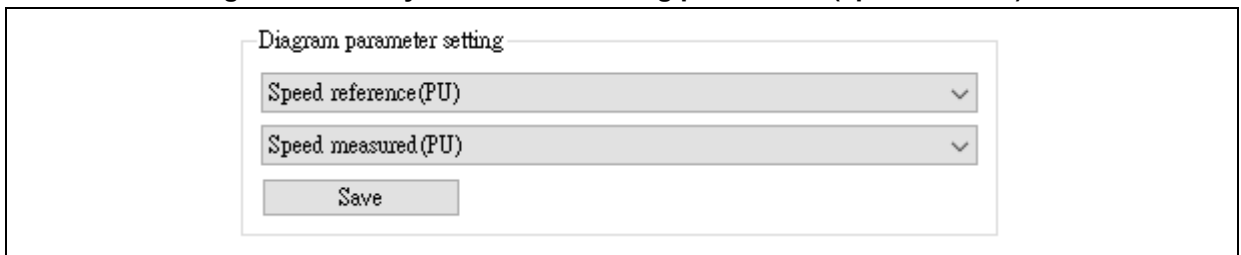


Diagram parameter setting

Speed reference(PU) ▼

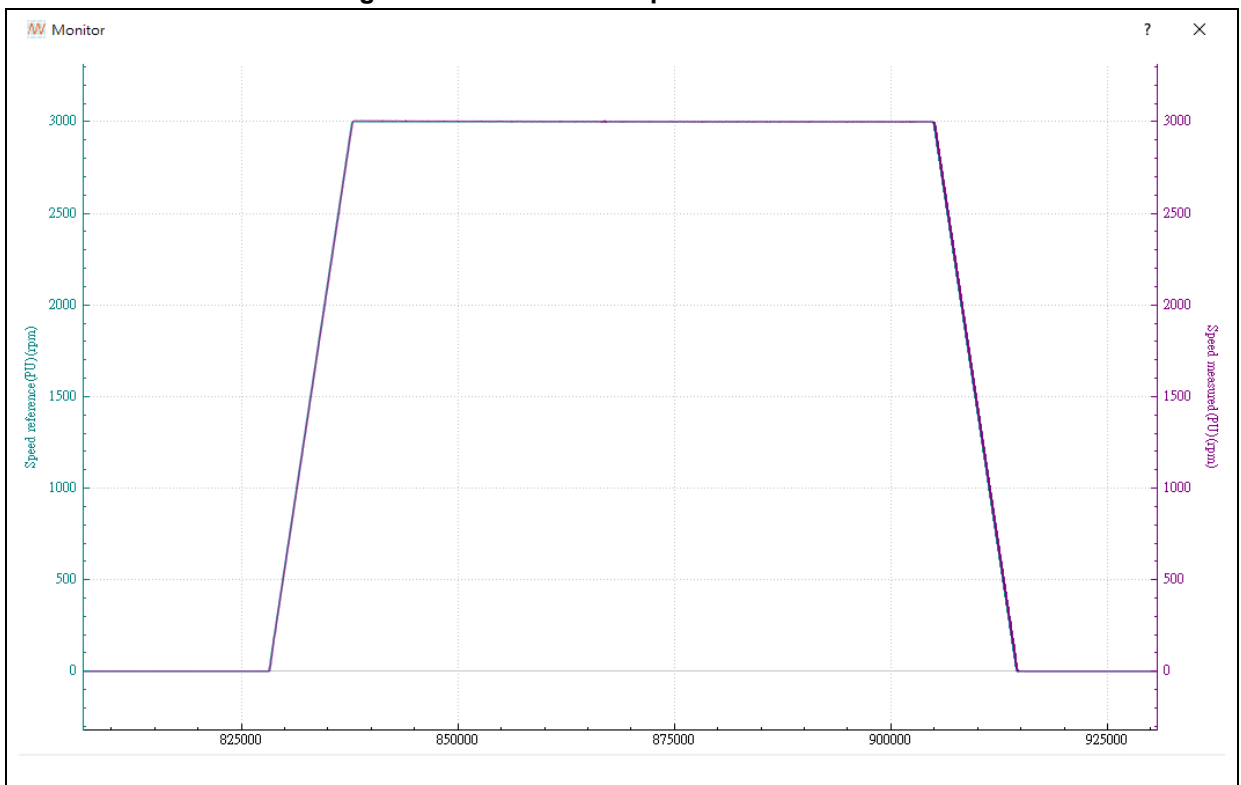
Speed measured(PU) ▼

Save

STEP-6: Click the drawing icon to open the waveform window.

STEP-7: Check whether the speed response is as expected, as shown in Figure 47. If it is not as expected, click to stop the motor and repeat STEP-2~STEP-7.

Figure 47. Waveform in speed control mode

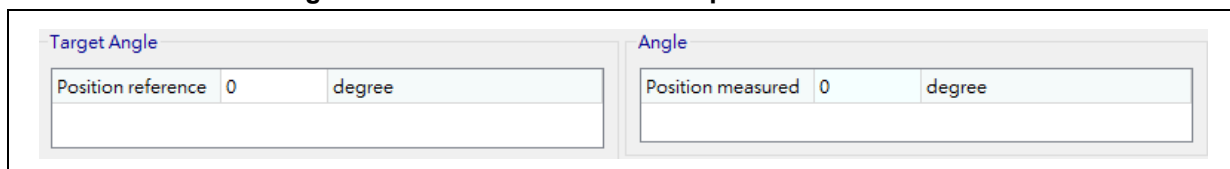


2.4.8 Position control

In position control mode, users can adjust the position PID parameters, and check the response after adjustment through waveform drawing.

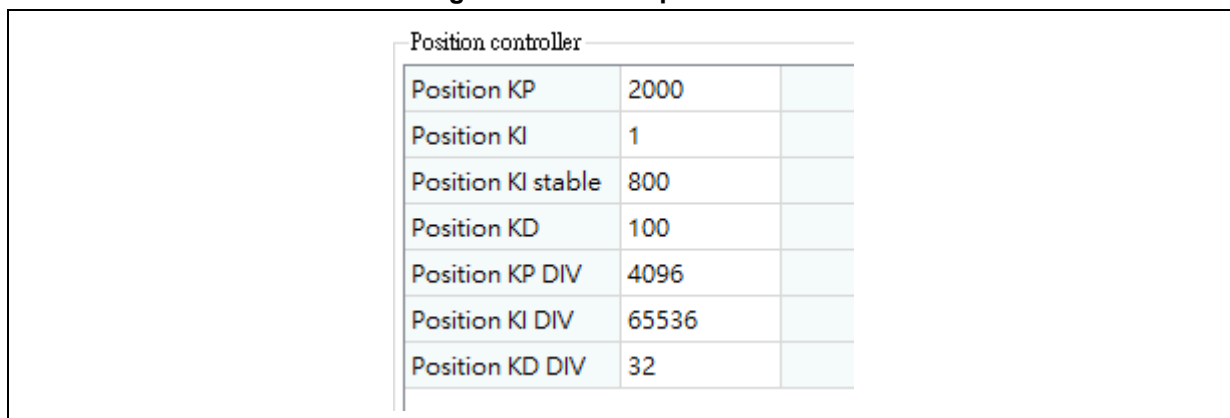
STEP-1: Select “Position Control”. With the encoder aligned, the motor rotor angle after rotation is not at the zero position exactly. In this case, both position reference and position measured are the angle after motor rotation.

Figure 48. Position reference and position measured



STEP-2: Set position PID parameters.

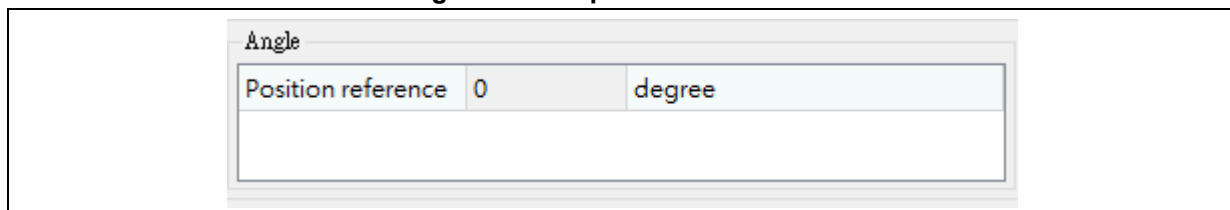
Figure 49. Set PID parameters



Position controller		
Position KP	2000	
Position KI	1	
Position KI stable	800	
Position KD	100	
Position KP DIV	4096	
Position KI DIV	65536	
Position KD DIV	32	

STEP-3: Set “Position reference”, for example, 3600 degrees.

Figure 50. Set position reference



STEP-4: Click on “Start Motor”.

STEP-5: Set “Position reference(PU)” and “Position measured(PU)” in “Diagram parameter setting”, and then click “Save”.

Figure 51. Modify channel monitoring parameters (position control)

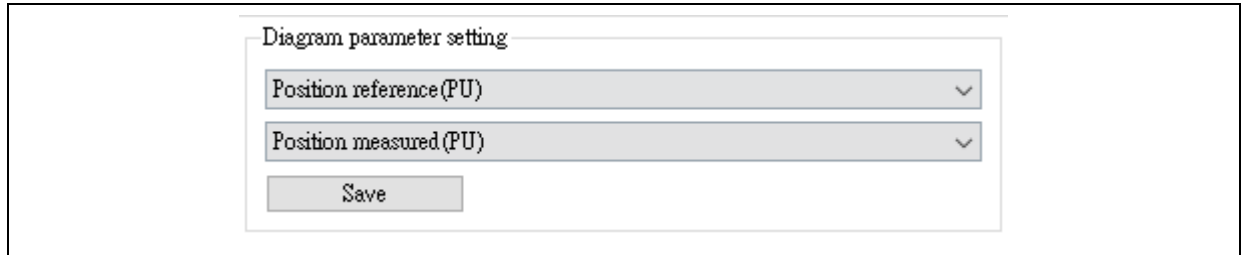


Diagram parameter setting

Position reference (PU) ▼

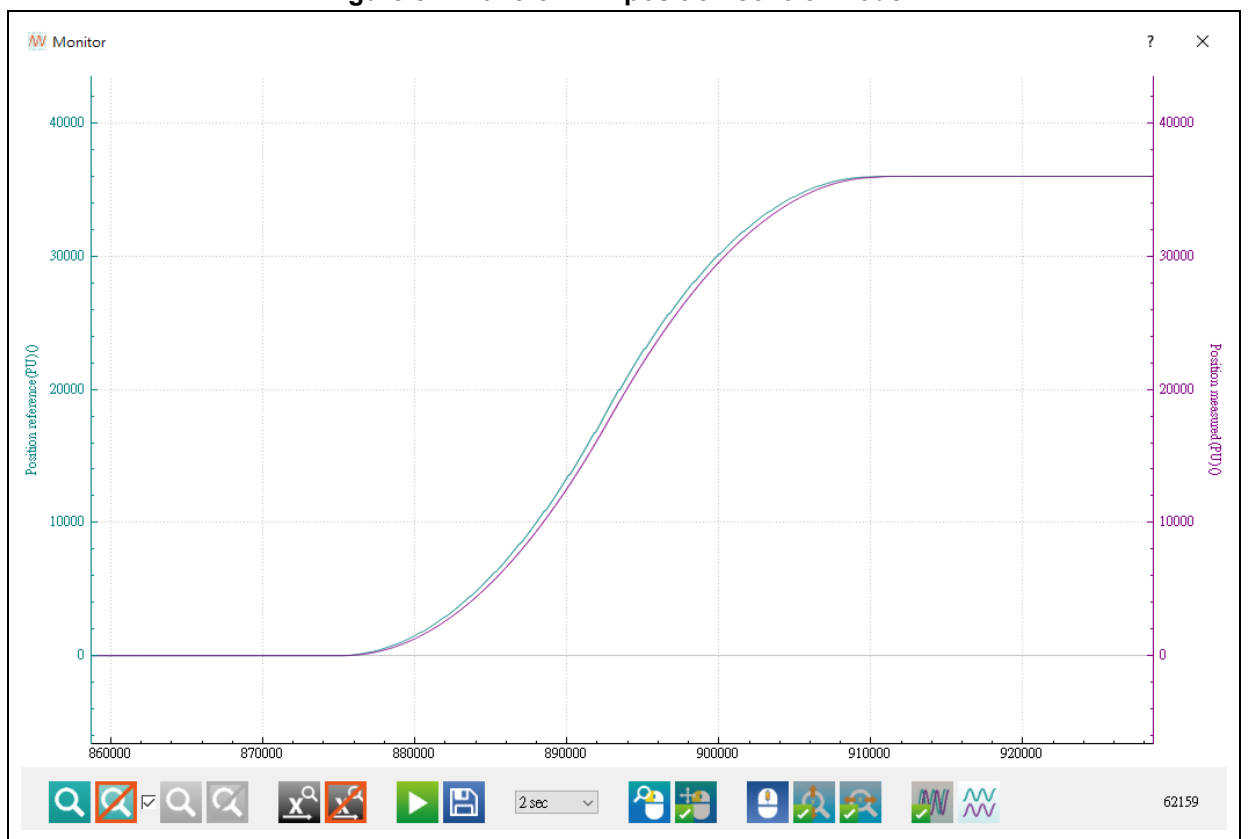
Position measured (PU) ▼

Save

STEP-6: Click the drawing icon to open the waveform window.

STEP-7: Check whether the position response is as expected, as shown in Figure 52. If it is not as expected, click to stop the motor and repeat STEP-2~STEP-7.

Figure 52. Waveform in position control mode



2.4.9 Speed voltage control

The speed voltage control method is used to control output voltage according to the speed error. It is required when the WITHOUT_CURRENT_CTRL (voltage control only) is enabled or the low-speed range of LOW_SPEED_VOLT_CTRL (low-speed voltage control) is enabled.

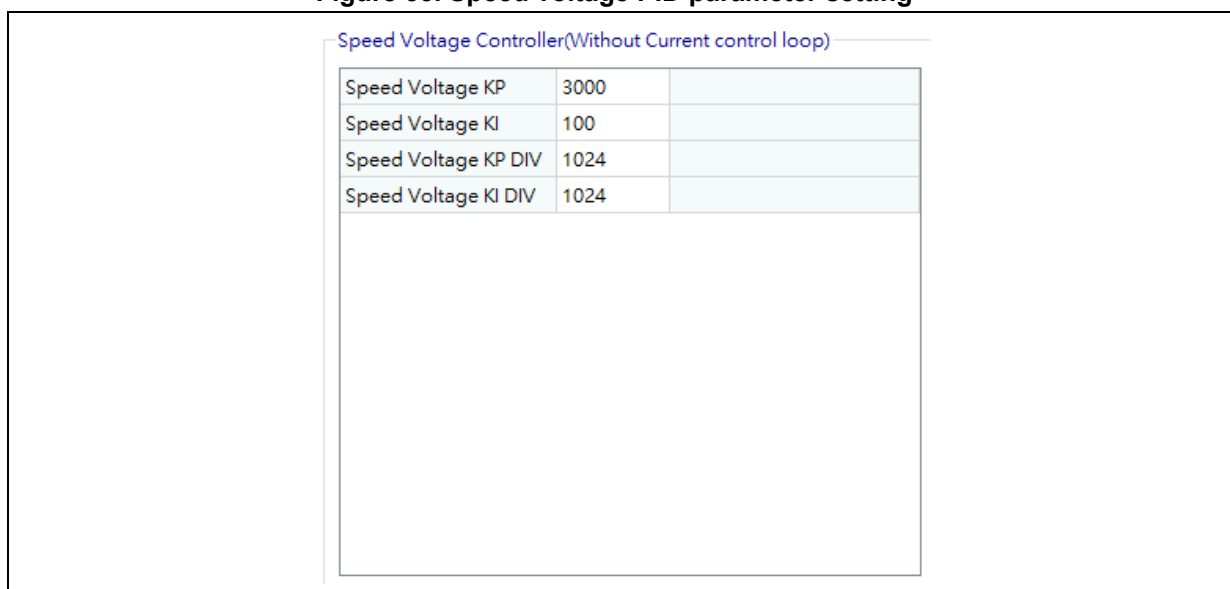
In this mode, users can set the speed voltage PID parameters, and check the response through waveform.

Follow the below steps.

STEP-1: Select “Speed Control”.

STEP-2: Set speed voltage PID parameters, acceleration and deceleration.

Figure 53. Speed voltage PID parameter setting

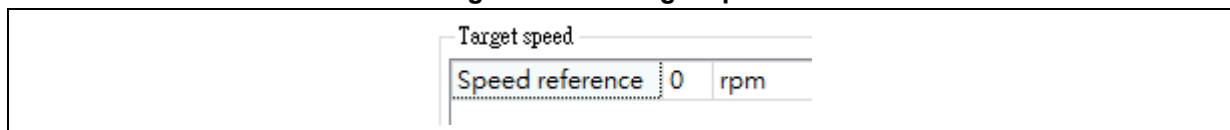


Speed Voltage Controller(Without Current control loop)

Speed Voltage KP	3000	
Speed Voltage KI	100	
Speed Voltage KP DIV	1024	
Speed Voltage KI DIV	1024	

STEP-3: Set “software control” as the control source and set the target speed (Speed reference).

Figure 54. Set target speed



Target speed

Speed reference 0 rpm

STEP-4: Click on “Start Motor” button.

STEP-5: Set Speed reference(PU) and Speed measured(PU) in “Diagram parameter setting” and then click on “Save”.

Figure 55. Set Speed reference(PU) and Speed measured(PU) (speed voltage control)

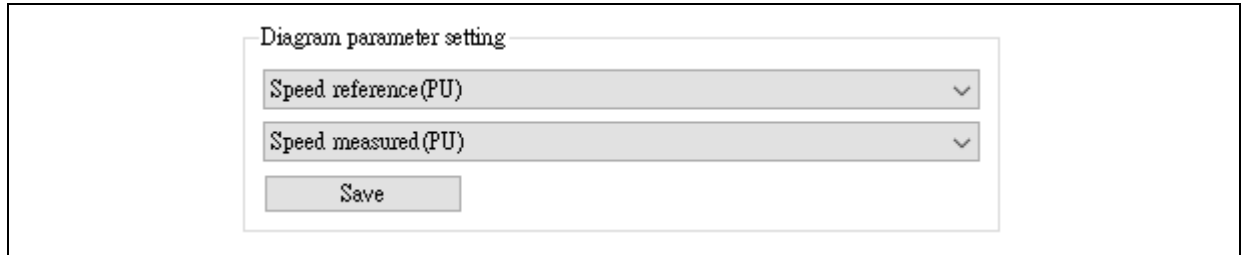


Diagram parameter setting

Speed reference(PU) ▼

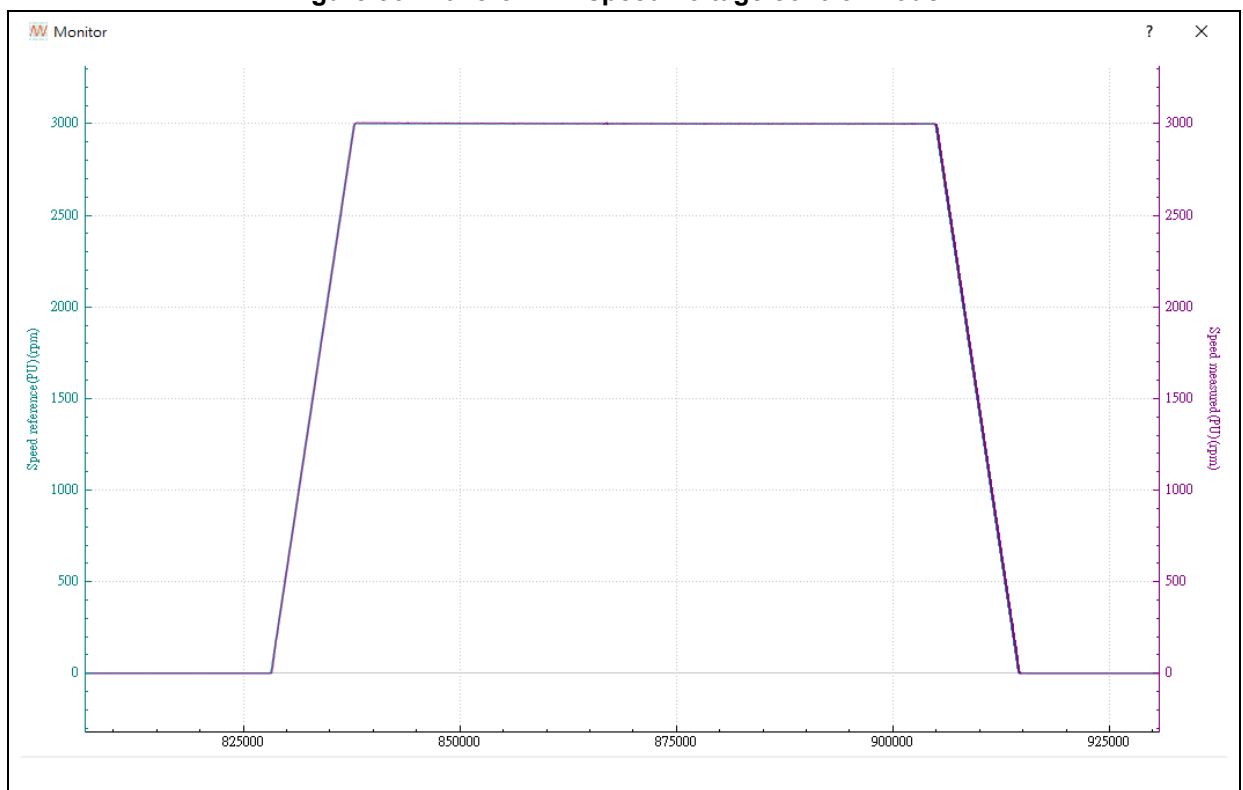
Speed measured(PU) ▼

Save

STEP-6: Click the drawing icon to open the waveform window.

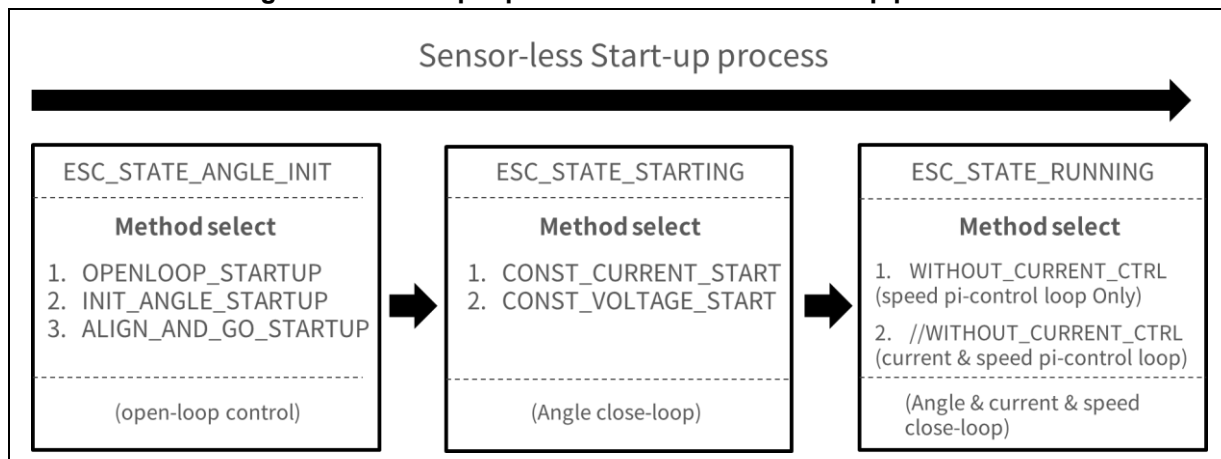
STEP-7: Check whether the speed response is as expected, as shown in Figure 53. If it is not as expected, click to stop the motor and repeat STEP-2~STEP-7.

Figure 56. Waveform in speed voltage control mode



2.4.10 Six-step square wave sensorless control

Figure 57. Six-step square wave sensorless startup process



Related parameters are listed in Table 2.

Table 2. Six-step square-wave sensorless control related parameters

ESC_STATE_ANGLE_INIT		
Method	Definition	Description
OPENLOOP_STARTUP	OLC_INIT_SPD	Open loop initial speed (unit: rpm)
	OLC_FINAL_SPD	Open loop final speed (unit: rpm)
	OLC_TIMES	Set the times from initial speed to rise up to final speed (unit: times)
	OLC_INIT_VOLT	Open loop initial voltage (unit: V)
	OLC_VOLT_INC	Open loop incremental voltage (unit: V)
	OLC_STARTUP_PERIOD	Set the period from open loop startup to closed loop (unit: ms) (for BLDC sensorless use only)
ALIGN_AND_GO_STARTUP	LOCK_VOLT	Rotor alignment voltage before startup (unit: V)
	LOCK_PERIOD	Rotor alignment time before startup (unit: ms)
INIT_ANGLE_STARTUP	ANGLE_INIT_DETECT_DUTY	PWM DUTY value detected at an initial angle (unit: PWM timing base)
	ANGLE_INIT_I_DIFF	Difference in initial angle detection

STARTING		
Method	Definition	Description
CONST_CURRENT_START	START_CURRENT	Current value for constant current startup (unit: ampere)
	SENSE_HALL_TIMES	Set phase-change times before open loop enter to closed loop (unit: times)
CONST_VOLTAGE_START	START_VOLTAGE	Voltage value for constant voltage startup (unit: V)
	SENSE_HALL_TIMES	Set phase-change times before open loop enter to closed loop (unit: times)

RUNNING		
Method	Definition	Description
Current control	PID_IS_KP_DIV_LOG	Bus current control proportional gain divisor (Q16 mode)
	PID_IS_KI_DIV_LOG	Bus current control integral gain divisor (Q16 mode)
	PID_IS_KP_DEFAULT	Bus current control proportional gain (Q15 mode)
	PID_IS_KI_DEFAULT	Bus current control integral gain (Q15 mode)
Speed control (high speed range in LOW_SPEED_VOLT_CTRL)	PID_SPD_KP_DIV_DIV	Speed control proportional gain divisor (Q16 mode)
	PID_SPD_KI_DIV_DIV	Speed control integral gain divisor (Q16 mode)
	PID_SPD_KP_DEFAULT	Speed control proportional gain (Q15 mode)
	PID_SPD_KI_DEFAULT	Speed control integral gain (Q15 mode)
Speed voltage control (low speed range in LOW_SPEED_VOLT_CTRL and WITHOUT_CURRENT_CTRL)	HYSTERESIS_LOW_SPEED	Minimum speed from current controller to voltage controller in low-speed voltage mode (rpm)
	HYSTERESIS_HIGH_SPEED	Maximum speed from voltage controller to current controller in low-speed voltage mode (rpm)
	PID_SPD_VOLT_KP_GAIN_DIV	Low-speed voltage controller proportional gain divisor (Q16 mode) (LOW_SPEED_VOLT_CTRL and WITHOUT_CURRENT_CTRL)
	PID_SPD_VOLT_KI_GAIN_DIV	Low-speed voltage controller integral gain divisor (Q16 mode) (LOW_SPEED_VOLT_CTRL and WITHOUT_CURRENT_CTRL)
	PID_SPD_VOLT_KP_DEFAULT	Low-speed voltage controller proportional gain (Q15 mode) (LOW_SPEED_VOLT_CTRL and WITHOUT_CURRENT_CTRL)
	PID_SPD_VOLT_KI_DEFAULT	Low-speed voltage controller integral gain (Q15 mode) (LOW_SPEED_VOLT_CTRL and WITHOUT_CURRENT_CTRL)

Different from sensed control mode, the sensorless control detects BEMF to estimate the rotor position. Users need to set Start Current or Start Voltage at startup according to the motor configurations, as shown in Figure 58.

Figure 58. Six-step sensorless control parameter setting (BEMF detection with comparator)

Sensor-less control(six-step)		
Start Current	0.799	A
Start Voltage	0.087	V

1. Start Current/Start Voltage (alternative)

- a) Start current: This is the initial current value for sensorless control during startup, measured in amperes (A).
- b) Start voltage: This is the initial voltage value for sensorless control during startup, measured in volts (V).

* Set the start current or start voltage, dependent on constant current startup or constant voltage startup.

If the method of detecting back-EMF commutation using an ADC needs adjustment, there are four parameters that can be tuned:

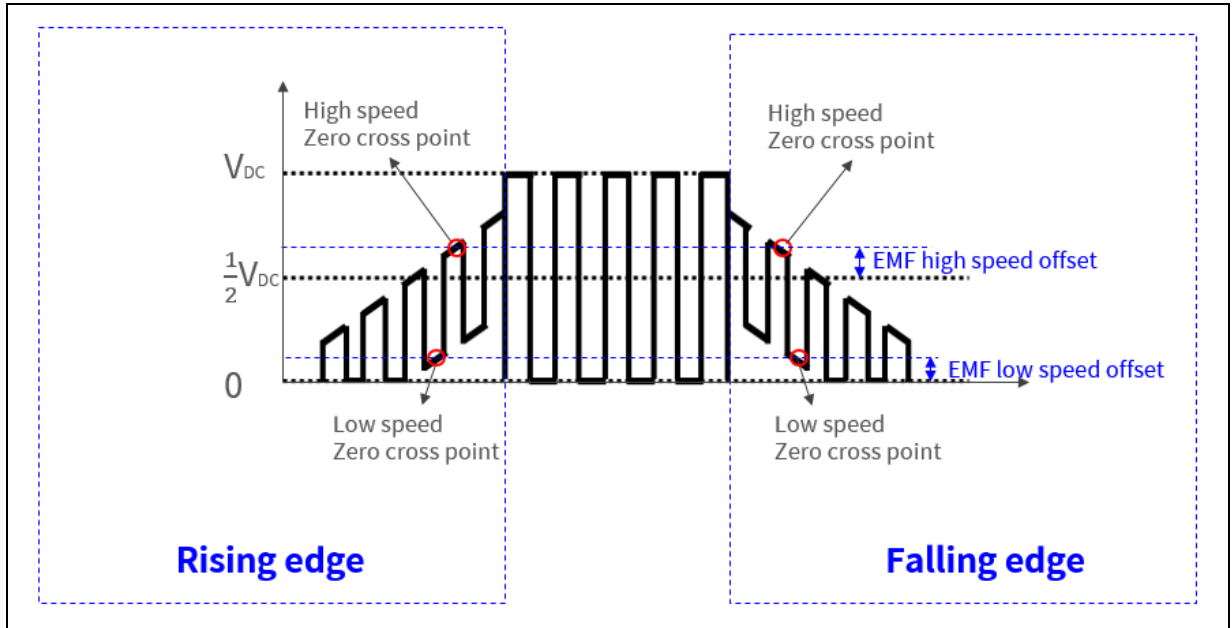
- EMF low speed offset (rising)
- EMF low speed offset (falling)
- EMF high speed offset (rising)
- EMF high speed offset (falling)

As shown in Figure 59 and Figure 60, the correct commutation point is achieved when the back-EMF signal is symmetrical, as illustrated in Figure 61.

Figure 59. Six-step sensorless control parameter setting (BEMF detection with ADC)

Sensor-less control(six-step)		
Start Current	0.399	A
Start Voltage	0.000	V
EMF low speed offset(Rising)	35	
EMF low speed offset(Falling)	35	
EMF high speed offset(Rising)	-100	
EMF high speed offset(Falling)	-100	

Figure 60. BEMF parameters (BEMF detection with ADC)



1. EMF low speed offset(rising), EMF low speed offset(falling) (note [3])

In low speed mode, detect the BEMF zero-crossing point during PWM OFF period. The “EMF low speed offset” is the level for BEMF zero-crossing point detection (positive edge: rising, negative edge: falling). If the level is set to a positive value (or negative value), the detection threshold moves upwards (or downwards) relative to 0.

It is the configuration for AT32 MOTOR EVB Board, which needs to be modified according to the board being used. In this example, the rising is 35 and the falling is 35, unit: ADC digital readout of BEMF feedback voltage.

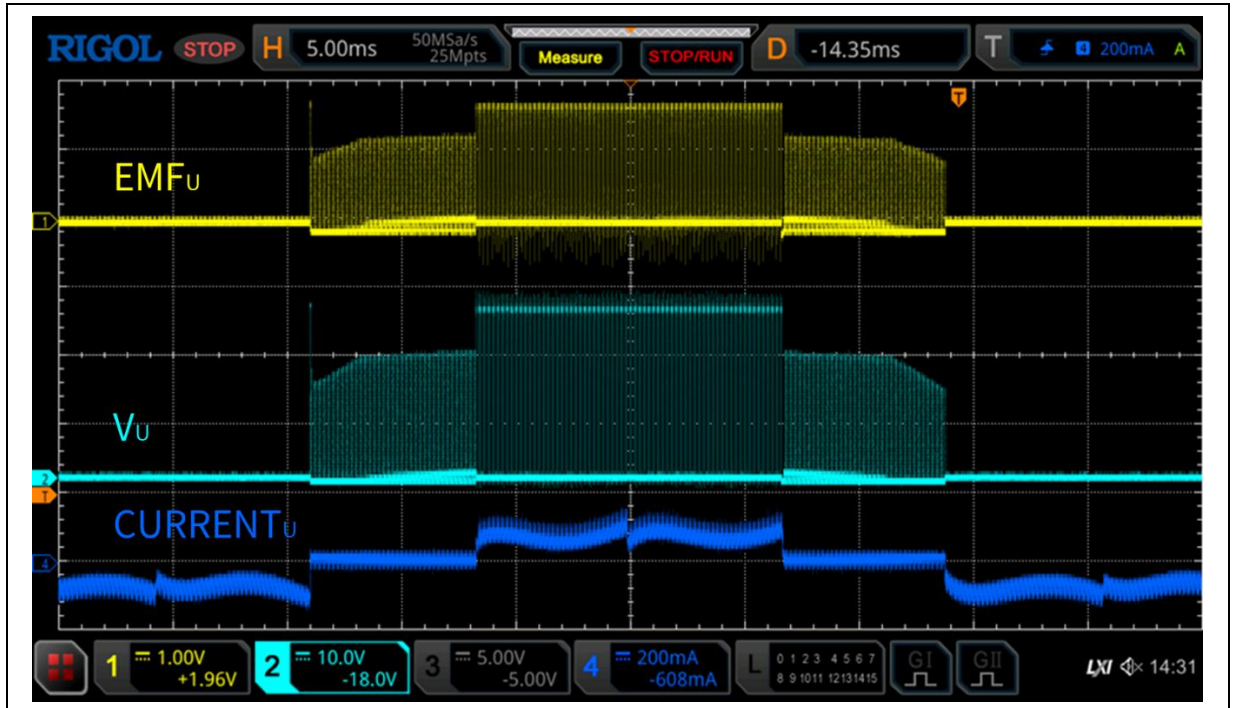
2. EMF high speed offset(rising), EMF high speed offset(falling) (note [3])

In high speed mode, detect the BEMF zero-crossing point during PWM ON period. The “EMF high speed offset” is the level for BEMF zero-crossing point detection (positive edge: rising, negative edge: falling). If the level is set to a positive value (or negative value), the detection threshold moves upwards (or downwards) relative to $1/2 V_{DC}$.

It is the configuration for AT32 MOTOR EVB Board, which needs to be modified according to the board being used. In this example, the rising is rising is -100 and the falling is -100, unit: ADC digital readout of BEMF feedback voltage.

Note [3]: No need to modify for BEMF detection with comparator. These parameters values are related to the hardware circuit of the drive board. Keep the above configurations for AT32 MOTOR EVB Board, and do not change.

Figure 61. Back-EMF Symmetrical Waveform Diagram



2.4.11 Sensorless FOC control

Different from control with sensors, the sensorless control detects BEMF to estimate the rotor position. The AT motor library uses Luenberger observer and Q-PLL method for BEMF estimation. Parameters are as shown in Figure 62, and users can use open loop control to run the motor to tune these parameters.

Figure 62. Sensorless observer + PLL parameters

Sensor-less observer+PLL	
Observer C1	15000
Observer C2	-20000
PLL KP	5000
PLL KI	5
PLL KP DIV	32768
PLL KI DIV	32768

In addition, AT motor library provides three startup modes. Related macro definitions (motor_control_drive_param.h) are listed in Table 3, and the corresponding parameters in the host are shown in Figure 63 and Figure 64.

Table 3. FOC sensorless control related parameters

Sensorless startup mode	Definition	Description
OPENLOOP_STARTUP	STARTUP_CURRENT	Initial current command for open loop control (unit: ampere)
	STARTUP_MAX_SPD	Open loop maximum speed before entering to closed loop (unit: rpm)
	STARTUP_OL_SLOPE	Open loop startup acceleration (unit: rpm/s)
ALIGN_AND_GO_STARTUP	STARTUP_CURRENT	Initial current command for align-and-go startup(unit: ampere)
	STARTUP_ALIGN_TIME	Rotor alignment time before startup (unit: ms)
	STARTUP_START_TIME	Startup time after rotor alignment (entering closed loop) (unit: ms)
INIT_ANGLE_STARTUP	DETECT_PULSE_WIDTH	Voltage pulse width for rotor initial angle detection (unit: us)
	STARTUP_CURRENT	Initial current command for open loop control (unit: ampere)
	STARTUP_MAX_SPD	Open loop maximum speed before entering to closed loop (unit: rpm)
	STARTUP_OL_SLOPE	Open loop startup acceleration (unit: rpm/s)

Figure 63. Open loop startup and initial angle detection startup

Start up		
Startup Start Current	0.499	A
Startup Max. Speed	400	rpm
Startup Open Loop Slope	800	rpm/s

Figure 64. Align-and-go startup

Start up		
Startup Start Current	0.499	A
Startup Align Time	1000	ms
Satrtup Start Time	10	ms

2.5 Auto identification and tuning of parameters

Automatic parameter identification is divided into two parts, namely automatic identification of motor coil parameters and self-tuning of current PI controller parameters.

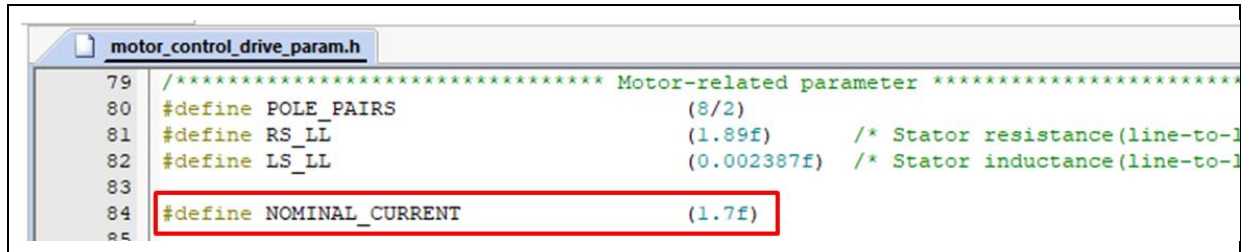
Auto identification of motor winding parameters are not required if there are RS_LL and LS_LL parameters. Users can directly modify macro definitions of RS_LL and LS_LL in *motor_control_drive_param.h* file.

Note [1]: These parameters can also be obtained by auto identification of motor winding parameters, as detailed in Section 2.5.1.

2.5.1 Auto identification of motor winding parameters

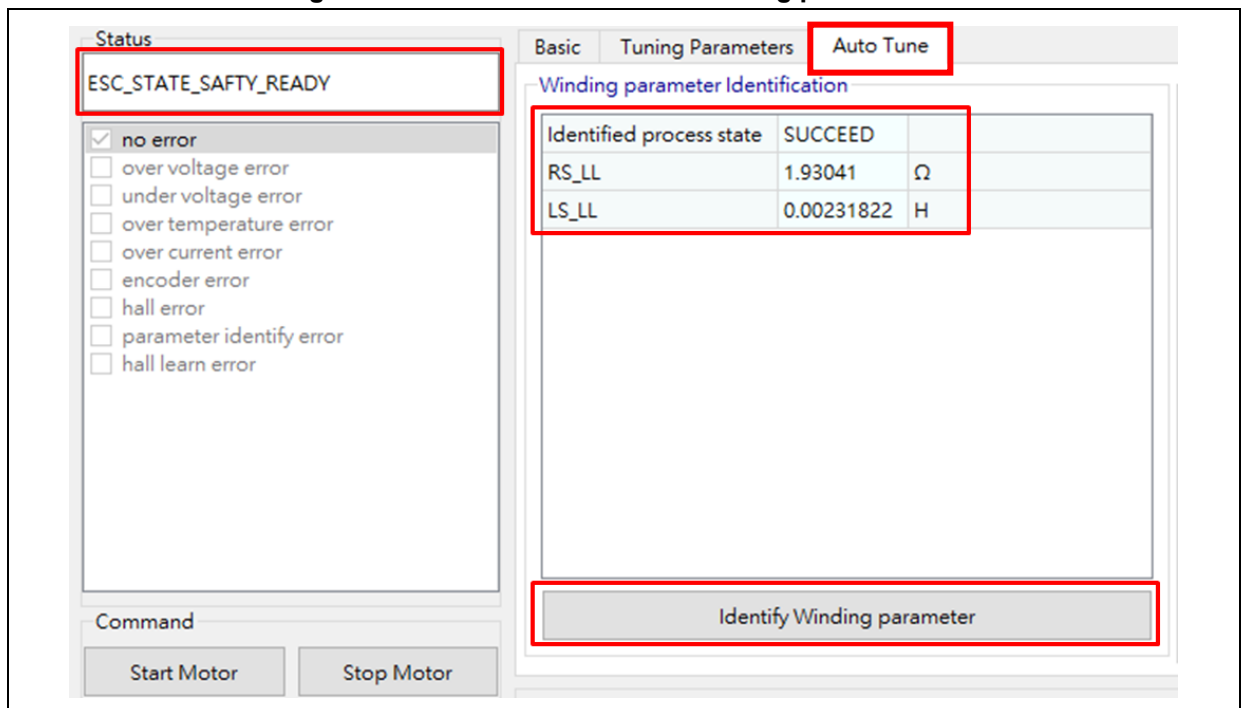
STEP-1: Fill the macro definition NOMINAL_CURRENT into *motor_control_drive_param.h* file, and then recompile and re-program code, as shown in Figure 65.

Figure 65. Macro definition of motor nominal current



STEP-2: Go to UI --> Auto Tune, and click the “Identify Winding parameter” button in “ESC_STATE_SAFETY_READY” status to perform winding parameter identification, as shown in Figure 66.

Figure 66. Auto identification of winding parameters



STEP-3: Check and confirm “SUCCEED” of “Identified process state”; get the RS_LL and LS_LL and fill them into the corresponding macro definitions in *motor_control_drive_param.h* file, as shown in Figure 67.

Figure 67. Macro definitions of winding parameters

```

motor_control_drive_param.h
89
90 /***** Motor-related parameter *****/
91 /* Motor parameters */
92 #define RS_LL          (1.89)          /* Stator resistance, ohm */
93 #define LS_LL          (0.002387)     /* Stator inductance, H */

```

2.5.2 Current PI controller parameter auto tuning

STEP-1: Fill in the micro definition of input DC voltage (VDC_RATED) into motor_control_drive_param.h file, and recompile and re-program, as shown in Figure 68.

Figure 68. Motor rated voltage

```

motor_control_drive_param.h
100
101 /***** Drive-related parameter *****/
102 /* basic */
103 #define VDC_RATED      (24.0f)
104 #define V_SENSE_GAIN   (10/(3.9f+180+10)) // 0.05157
105 #define ADC_REFERENCE_VOLT (3.3f)
106 #define ADC_DIGITAL_SCALE_12BITS (4095)

```

STEP-2: Go to Auto Tune window, and click the “Auto-tune Current PI parameter” button in “safety ready” status to perform auto tuning of current PI controller parameters. After the auto tuning is completed, current Q axis and D axis PI controller parameters are automatically updated, as shown in Figure 69.

Figure 69. Current PI controller parameter auto tuning

Auto-tune PI parameter of Current controller

Torque KP	7717
Torque KI	6110
Torque KP DIV	512
Torque KI DIV	8192

Auto-tune Current PI parameter

STEP-3: Fill the parameters shown in Figure 69 into the corresponding macro definition (motor_control_drive_param.h), then recompile and re-program code, as shown in Figure 70.

Figure 70. Macro definition of current PI controller parameters

```

motor_control_drive_param.h
242  /* pi parameter */
243  #define PID_IS_KP_DEFAULT      2000
244  #define PID_IS_KI_DEFAULT      200
245  #define PID_IS_KP_DIV          256
246  #define PID_IS_KP_DIV_LOG      LOG2(PID_IS_KP_DIV)
247  #define PID_IS_KI_DIV          1024
248  #define PID_IS_KI_DIV_LOG      LOG2(PID_IS_KI_DIV)

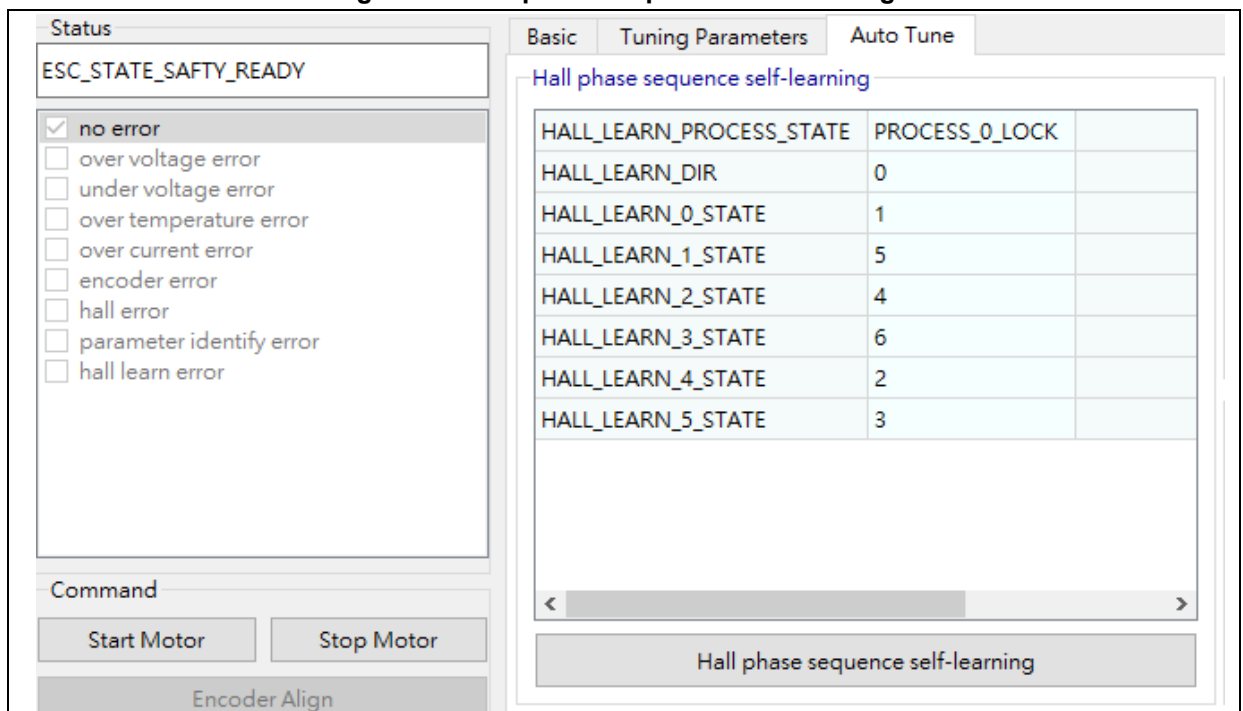
```

2.5.3 Hall phase sequence self-learning

STEP-1: Connect to UI interface.

STEP-2: Enter debug code and go to “Auto Tune” window; press “Hall phase sequence self-learning” button in ESC_STATE_SAFETY_READY status to execute Hall phase sequence self-learning, as shown in Figure 71.

Figure 71. Hall phase sequence self-learning



STEP-3. The motor runs in open loop mode. If the motor is running in the wrong direction, wait for the motor to stop and then press the “Hall phase sequence self-learning” button again to reverse the direction. Self-learning is complete when the motor stops.

Note: If the motor does not run smoothly or does not run at all, go back to the open loop control to adjust the open loop voltage and speed, then recompile and re-program, and restart STEP-2.

STEP-4: Check whether the HALL_LEARN_PROCESS_STATE is “PROCESS_4_FINISH”.

STEP-5: Fill in the values of “HALL_LEARN_DIR” and “HALL_LEARN_0_STATE~HALL_LEARN_5_STATE” to the “hall learn table” macro definition

(motor_control_drive_param.h) (Figure 72), and then recompile and re-program code.

Figure 72. Macro definition of Hall self-learning

Basic Tuning Parameters Auto Tune

Hall phase sequence self-learning

HALL_LEARN_PROCESS_STATE	PROCESS_0_LOCK
HALL_LEARN_DIR	0
HALL_LEARN_0_STATE	1
HALL_LEARN_1_STATE	5
HALL_LEARN_2_STATE	4
HALL_LEARN_3_STATE	6
HALL_LEARN_4_STATE	2
HALL_LEARN_5_STATE	3

Hall phase sequence self-learning

```

98
99 /* hall learn table */
100 #if defined (SENSORLESS)
101 #define HALL_LEARN_DIR (0)
102 #define HALL_LEARN_0_STATE (2)
103 #define HALL_LEARN_1_STATE (3)
104 #define HALL_LEARN_2_STATE (1)
105 #define HALL_LEARN_3_STATE (5)
106 #define HALL_LEARN_4_STATE (4)
107 #define HALL_LEARN_5_STATE (6)
108 #else
109 #define HALL_LEARN_DIR (0)
110 #define HALL_LEARN_0_STATE (1)
111 #define HALL_LEARN_1_STATE (5)
112 #define HALL_LEARN_2_STATE (4)
113 #define HALL_LEARN_3_STATE (6)
114 #define HALL_LEARN_4_STATE (2)
115 #define HALL_LEARN_5_STATE (3)
116 #endif
  
```

3 Revision history

Table 4. Document revision history

Date	Version	Revision note
2022.11.18	2.0.1	Initial release.
2022.12.01	2.0.2	Optimized descriptions.
2023.01.06	2.0.3	Added current control, and parameters relevant to sensorless FOC control
2023.03.02	2.0.4	Modified relevant parameters in six-step square-wave sensorless control.
2023.04.20	2.1.0	Updated instructions to the new version; Added Section 2.4.8 "Position control".
2023.10.05	2.1.1	Updated operation instructions.
2024.02.29	2.1.2	Added instructions to the automatic identification of motor coil parameters and self-tuning of current PI controller parameters.
2024.05.24	2.1.3	Added instructions to the Hall learn feature, and updated instructions to six-step square wave sensorless control.
2024.12.27	2.1.4	Deleted parameter update icon, and added Section 2.4.9 "speed voltage control".
2025.04.11	2.1.5	Added instructions to the waveform storage feature.
2025.10.09	2.1.6	Add ADC detection for back-EMF adjustment description

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